

# Spatiotemporal Calibration of Doppler Velocity Logs for Underwater Robots

Hongxu Zhao<sup>1</sup>, Guangyang Zeng<sup>2</sup>, Yunling Shao<sup>1</sup>, Tengfei Zhang<sup>1</sup>, Yuzhe Li<sup>3</sup>, and Junfeng Wu<sup>1</sup>

**Abstract**—Accurate fusion of Doppler Velocity Logs (DVLs) with other sensors requires both spatial extrinsics and temporal alignment. However, existing underwater DVL calibration methods often assume known lever arms and synchronized clocks, and rarely estimate full calibration parameters jointly. We propose Unified Iterative Calibration (UIC), a spatiotemporal calibration framework for a DVL and a generic co-sensor that can provide a continuous-time motion estimate. We model the base motion with a continuous-time Gaussian process, enabling velocity queries at time-shifted DVL timestamps. UIC computes a sequential initialization: (i) it estimates the clock offset and DVL scale using low-angular-rate segments via velocity-norm matching, and (ii) estimates the DVL-to-base rotation and lever arm through a covariance-corrected closed-form stacked regression with  $SO(3)$  projection. A nonlinear refinement further reduces residuals. Simulations with DVL outliers and pool experiments demonstrate robustness and high accuracy downstream odometry estimation.

## I. INTRODUCTION

Doppler Velocity Logs (DVLs) are key sensors for underwater navigation, which can provide body-frame velocity measurements [1]–[5]. For multi-sensor fusion (e.g., DVL+camera/SINS), accurate calibration requires both the spatial extrinsics and temporal alignment, since even a small constant clock offset can cause systematic velocity mismatch under dynamic motion.

Underwater DVL calibration has been studied mainly for DVL–camera systems and for standalone DVLs with navigation-grade reference velocity sources. Westman et al. [6] and Xu et al. [7] formulated DVL–camera calibration as trajectory alignment, but typically rely on known DVL–IMU extrinsics and do not estimate a lever arm and a constant time offset jointly. For IMU-free DVLs, works such as [8]–[11] estimate rotation and scale using SINS/GPS-type velocity references, which limits applicability when the co-sensor does not directly provide body-frame velocities. Temporal calibration for heterogeneous sensors has been widely studied on land by fitting continuous-time trajectories and querying measurements at shifted timestamps (e.g., B-splines/GPs) [12]–[16], but has been less explored for DVLs.

We address DVL spatiotemporal calibration by estimating  $\Theta = \{^b\delta_{T,d}, \kappa_d, {}^dR_b, {}^d\mathbf{t}_b\}$ , where  ${}^b\delta_{T,d}$  is a

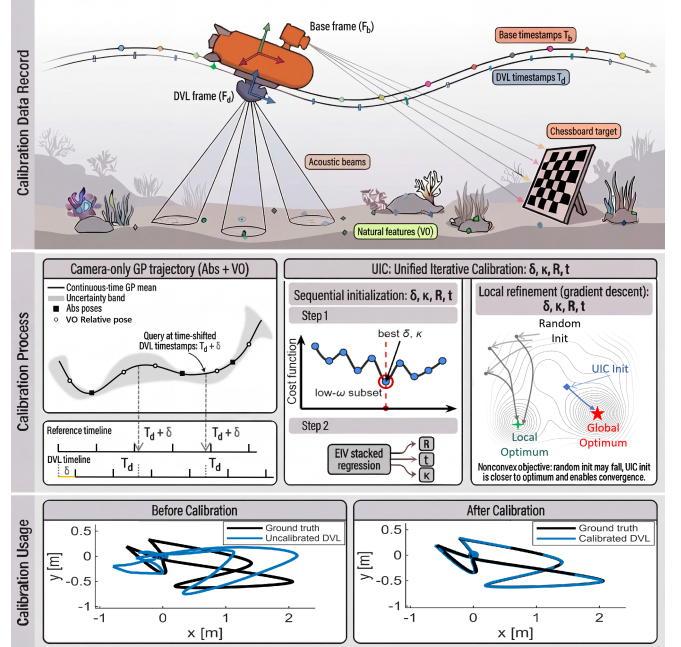


Fig. 1: UIC overview: (1) infer a continuous-time base motion with a GP prior, (2) initialize clock offset/scale using low-rotation segments, (3) Initialize extrinsics via closed-form stacked regression, and (4) optionally refine  $\Theta$  by local nonlinear least squares.

constant clock offset (base time minus DVL time),  $\kappa_d$  is the DVL velocity scale factor, and  $({}^dR_b, {}^d\mathbf{t}_b)$  are the rigid extrinsics. Our formulation does not require the co-sensor to output velocities; it only needs to support a continuous-time motion estimate from which base-frame linear and angular velocities can be queried.

Our main contributions are:

- An EIV spatiotemporal calibration model with large-sample characterization under idealized assumptions. We cast DVL spatiotemporal calibration as an error-in-variables (EIV) nonlinear least-squares problem under a continuous-time GP motion prior, jointly estimating  $\Theta = \{^b\delta_{T,d}, \kappa_d, {}^dR_b, {}^d\mathbf{t}_b\}$  via GP-based time-shifted velocity querying. Under idealized modeling assumptions and sufficiently exciting motion, we characterize the large-sample behavior of the proposed sequential estimators, establishing bias elimination for all calibration variables.
- A decoupled calibration pipeline with principled sequential initialization and closed-form extrinsics + numerical refinement. We explicitly decouple the

<sup>1</sup>The School of Data Science, Chinese University of Hong Kong, Shenzhen, Shenzhen, P. R. China. hxsimon99@gmail.com, yunlingshao@link.cuhk.edu.cn, zhangtengfei@cuhk.edu.cn, junfengwu@cuhk.edu.cn. <sup>2</sup>The School of Intelligence Science and Engineering, Harbin Institute of Technology, Shenzhen, Shenzhen 518055, P. R. China. zengguangyang@hit.edu.cn. <sup>3</sup>The State Key Laboratory of Synthetical Automation for Process Industries, Northeastern University, Shenyang, P. R. China. yuzheli@mail.neu.edu.cn.

# 水下机器人的多普勒速度计时空校准

赵洪旭<sup>1</sup>, 曾广洋<sup>2</sup>, 邵云岭<sup>1</sup>, 张腾飞<sup>1</sup>, 李禹喆<sup>3</sup>, 吴俊锋<sup>1</sup>

摘要——精确地将多普勒速度计（DVL）与其他传感器融合，需要同时考虑空间外参和时间对齐。然而，现有的水下DVL校准方法通常假设已知支臂和同步时钟，很少联合估计完整的校准参数。我们提出了统一迭代校准（UIC），这是一种用于DVL和能够提供连续时间运动估计的通用协同传感器的时空校准框架。我们使用连续时间高斯过程来建模基座运动，从而能够在时间偏移的DVL时间戳处进行速度查询。UIC计算一个顺序初始化：（i）利用低角速率段通过速度范数匹配估计时钟偏移和DVL尺度；（ii）通过协方差校正的闭式堆叠回归及SO(3)投影估计DVL到基座的旋转和平移臂。非线性优化进一步减少残差。带有DVL异常值的仿真和水池实验显示

## 一、引言

多普勒速度测量仪（DVL）是水下导航的关键传感器，可以提供机体坐标系下的速度测量 [1] – [5]。对于多传感器融合（例如 DVL+摄像头/SINS），精确的标定需要同时考虑空间外参和时间对齐，因为即便是很小的恒定时钟偏差，在动态运动下也会导致系统性的速度不匹配。

水下DVL校准主要针对DVL-相机系统以及具有导航级参考速度源的独立DVL进行研究。Westman 等人 [6] 和 Xu 等人 [7] 将DVL-相机校准表述为轨迹对齐，但通常依赖已知的DVL-IMU外参，并且通常不同时估计力臂和恒定时间偏移。对于无IMU的DVL，诸如 [8]–[11] 的工作使用SINS/GPS类型的速度参考来估计旋转和尺度，这在共传感器不能直接提供机体坐标系速度时限制了适用性。异构传感器的时间校准在陆地上已被广泛研究，通过拟合连续时间轨迹并在移位时间戳处查询测量值（如B样条/高斯过程）[12]–[16]，但在DVL上研究较少。

我们通过估计  $\Theta = \{^b\delta_{T,d}, \kappa_d, {}^dR_b, q_b\}$  来解决 DVL 时空校准问题，其中  ${}^b\delta_{T,d}$  是一个

<sup>1</sup>香港中文大学（深圳）数据科学学院，中国深圳。hxzsimon99@gmail.com, yunlingshao@link.cuhk.edu.cn, zhangtengfei@cuhk.edu.cn, junfengwu@cuhk.edu.cn. <sup>2</sup>哈尔滨工业大学（深圳）智能科学与工程学院，中国深圳 518055. zengguangyang@hit.edu.cn. <sup>3</sup>东北大学过程工业综合自动化国家重点实验室，中国沈阳。yuzheli@mail.neu.edu.cn.

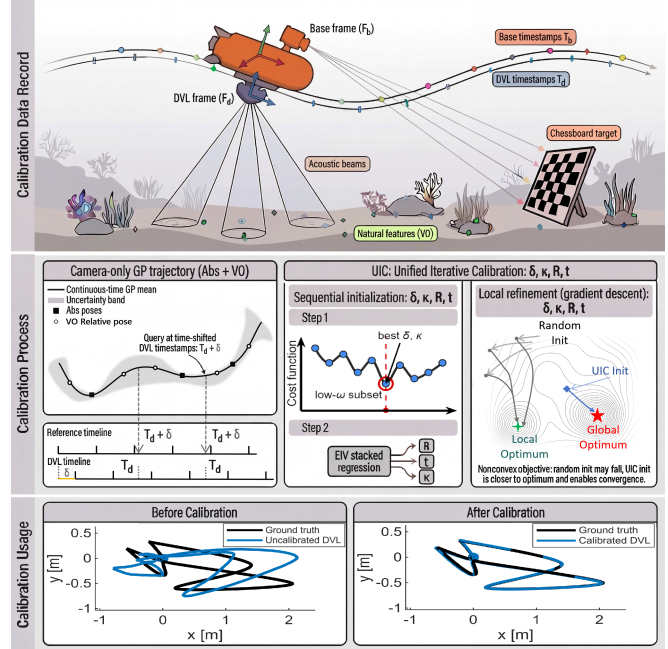


图 1: UIC 概览：（1）使用 GP 先验推断连续时间基准运动，（2）使用低旋转段初始化时钟偏移/比例，（3）通过闭式堆叠回归初始化外参，以及（4）可选择通过局部非线性最小二乘法优化  $\Theta$ 。

恒定时钟偏移（基准时间减去DVL时间）， $\kappa_d$ 是DVL速度尺度因子， $({}^dR_b, q_b)$ 是刚性外参。我们的公式不要求协同传感器输出速度；它只需要支持连续时间运动估计，从中可以查询基坐标系的线速度和角速度。

我们的主要贡献是：

- 在理想化假设下具有大样本特性的EIV时空校准模型。我们将DVL时空校准表述为在连续时间GP运动先验下的误差变量（EIV）非线性最小二乘问题，通过基于GP的时间偏移速度查询联合估计  $\Theta = \{^b\delta_{T,d}, \kappa_d, {}^dR_b, q_b\}$ 。在理想化建模假设和充分激励的运动条件下，我们描述了所提出的序贯估计器的大样本行为，建立了对所有校准变量的偏差消除。
- 一个具有原则性顺序初始化和封闭形式外参 + 数值优化的解耦校准管线。我们明确地解耦了

clock offset  ${}^b\delta_{T,d}$  from the spatial extrinsics to reduce mutual interference in joint optimization: leveraging (i) the rotation-invariance of velocity norms and (ii) near-zero angular-rate segments ( $\omega_b \approx 0$ ) to suppress the lever-arm term, we initialize  $({}^b\delta_{T,d}, \kappa_d)$  with minimal dependence on  $({}^dR_b, {}^dt_b)$ . Fixing the time offset, we then derive a bias-corrected closed-form estimator for the spatial extrinsics (with proper SO(3) projection), and use it as an initialization for optional finite-difference numerical-gradient descent refinement.

- FIM-based excitation analysis and trajectory design guidelines. We analyze the Fisher information matrix (FIM) of the proposed spatiotemporal calibration model to identify the motion excitations required to calibrate all parameters, and translate the analysis into practical trajectory requirements (e.g., combining low-rotation segments for temporal initialization with sufficiently rotational excitation for lever-arm estimation).

Notation: Throughout this paper, we adhere to the following notational conventions: the super-scripted  $(\cdot)^o$  represents the true value of a variable  $(\cdot)$ ,  $(\cdot)$  a measurement of  $(\cdot)$ , and  $(\hat{\cdot})$  an estimate to  $(\cdot)$ . For a vector  $a \in \mathbb{R}^3$ , let  $a^\wedge \in \mathbb{R}^{3 \times 3}$  be the skew-symmetric matrix that satisfies  $a \times b = a^\wedge b$  for all  $b \in \mathbb{R}^3$ , where  $\times$  is the vector cross product.

## II. Problem formulation

### A. DVL Measurement Model

When a DVL is mounted on an underwater robot, it emits acoustic beams and measures the Doppler shifts resulting from the robot's movement relative to its ambient environment, which enables the DVL to calculate a three-dimensional velocity under the assumption of a constant and known acoustic speed.

Mathematically, DVL measurements are resolved relative to its local body-fixed frame  $\{\mathcal{F}_d\}$ . To analyze the motion of the DVL with respect to a base frame  $\{\mathcal{F}_b\}$  (usually attached to the other sensor), let  $p_{o_d,b}$  denote the coordinates of the origin  $o_d$  of  $\{\mathcal{F}_d\}$  expressed in frame  $\{\mathcal{F}_b\}$ . Essentially, the DVL measures the linear velocity of  $o_d$  expressed in its own frame  $\{\mathcal{F}_d\}$ . Provided that the base frame and the DVL frame are rigidly connected, when the base moves with linear velocity  $v_b$  and angular velocity  $\omega_b$  (both expressed in  $\{\mathcal{F}_b\}$ ), the relationship between the DVL velocity  $v_d$  and  $(v_b, \omega_b)$  is

$$v_d = {}^dR_b(v_b + \omega_b \times {}^dt_b), \quad (1)$$

where  ${}^dR_b \in \text{SO}(3)$  is the rotation mapping coordinates from  $\{\mathcal{F}_b\}$  to  $\{\mathcal{F}_d\}$ , and  ${}^dt_b \in \mathbb{R}^3$ , which equals to  $p_{o_d,b}$ , is the lever arm from the base-frame origin to the DVL-frame origin expressed in  $\{\mathcal{F}_b\}$ .

To account for sound-speed mismatch (e.g., due to temperature and salinity) and other multiplicative scale biases, we include a positive scaling factor  $\kappa_d$ . We also

consider an additive measurement noise term. Hence, a real DVL measurement  $\hat{v}_d$  can be modeled as

$$\hat{v}_d = \kappa_d {}^dR_b(v_b + \omega_b \times {}^dt_b) + n_v, \quad (2)$$

where  $n_v \sim \mathcal{N}(0, Q_d)$  denotes the measurement noise whose covariance can be obtained from the DVL sensor.

### B. Unified Spatiotemporal Calibration Problem under GP Motion Prior

The robot motion relative to a fixed world frame is represented by a continuous-time state  $x(t)$  expressed in the base frame  $\{\mathcal{F}_b\}$ . We model  $x(t)$  as a Gaussian process (GP),

$$x(t) \sim \mathcal{GP}(\mu(t), K(t, t')), \quad (3)$$

where  $K(t, t') \triangleq \text{cov}(x(t), x(t'))$  is the (matrix-valued) GP covariance kernel (with chosen hyperparameters) [17]. The reference sensor suite attached to the base provides measurements  $\hat{\mathcal{M}}_b = \{\hat{m}_{b,j}\}_{j=1}^M$  time-stamped at  ${}^bT = \{{}^bT_j\}_{j=1}^M$ , with measurement model

$$\hat{m}_{b,j} = h(x({}^bT_j)) + n_{b,j}, \quad n_{b,j} \sim \mathcal{N}(0, R_{b,j}), \quad j = 1, \dots, M. \quad (4)$$

Define the stacked vector  $\hat{M}_b \triangleq [\hat{m}_{b,1}^\top \dots \hat{m}_{b,M}^\top]^\top$  and let  $H$  denote the Jacobian of  $h(\cdot)$  evaluated at the linearization point used by GP regression. Let  $X_b \triangleq [x({}^bT_1)^\top \dots x({}^bT_M)^\top]^\top$  be the stacked state at the base timestamps. Using  $\hat{M}_b$  and the GP prior, we compute a reference trajectory estimate (and its uncertainty) via GP regression, denoted by  $\hat{X}_b$  (and  $\Sigma_{X_b}$ ); the closed-form expressions are provided in Section III-B.

The DVL provides velocity measurements  $\hat{\mathcal{V}}_d = \{\hat{v}_{d,i}\}_{i=1}^N$  time-stamped by its local clock at  ${}^dT = \{{}^dT_i\}_{i=1}^N$ . A constant clock offset between the base and DVL clocks is modeled by  ${}^b\delta_{T,d}$ , such that the physical time of a DVL sample recorded at  ${}^dT_i$  corresponds to the base time  ${}^dT_i + {}^b\delta_{T,d}$ . The calibration variables of interest are collected in

$$\Theta \triangleq \{{}^b\delta_{T,d}, {}^dR_b, {}^dt_b, \kappa_d\},$$

where  ${}^dR_b \in \text{SO}(3)$  and  ${}^dt_b \in \mathbb{R}^3$  are the extrinsic parameters, and  $\kappa_d > 0$  is the DVL velocity scale factor. For a candidate  ${}^b\delta_{T,d}$ , define the aligned DVL timestamps in the base clock by  ${}^bT_{d,i}({}^b\delta_{T,d}) \triangleq {}^dT_i + {}^b\delta_{T,d}$ , and let  $X_d({}^b\delta_{T,d}) \triangleq [x({}^bT_{d,1}({}^b\delta_{T,d}))^\top \dots x({}^bT_{d,N}({}^b\delta_{T,d}))^\top]^\top$  denote the queried states at these aligned instants. Given  $\hat{X}_b$ , the queried mean  $\hat{X}_d({}^b\delta_{T,d})$  is obtained by GP interpolation (as an explicit function of  ${}^b\delta_{T,d}$ ), and the corresponding queried covariance  $\Sigma_{X_d}({}^b\delta_{T,d})$  can be computed as a byproduct of GP regression and interpolation following (8) and (9).

At aligned time  ${}^bT_{d,i}({}^b\delta_{T,d})$ , we expect the DVL measurement as  $\kappa_d {}^dR_b(\hat{v}_{b,i}({}^b\delta_{T,d}) + \hat{\omega}_{b,i}({}^b\delta_{T,d}) \times {}^dt_b)$ , where  $\hat{v}_{b,i}({}^b\delta_{T,d})$  and  $\hat{\omega}_{b,i}({}^b\delta_{T,d})$  are the linear and angular velocity components extracted from the queried state

为了减少联合优化中的相互干扰，将时钟偏移  ${}^b\delta_{T,d}$  从空间外参中分离出来：利用 (i) 速度模的旋转不变性，以及 (ii) 近零角速率段 ( $\omega_b \approx 0$ ) 来抑制杆臂项，我们以对  $({}^dR_b, {}^d t_b)$  依赖最小的方式初始化 ( ${}^b\delta_{T,d}, \kappa_d$ )。固定时间偏移后，我们随后导出用于空间外参的偏差校正闭式估计器（带有适当的 SO(3) 投影），并将其用作可选有限差分数值梯度下降优化的初始化。

- 基于FIM的激励分析和轨迹设计指南。我们分析所提出的时空校准模型的费舍尔信息矩阵（FIM），以确定校准所有参数所需的运动激励，并将分析结果转化为实际轨迹要求（例如，将用于时间初始化的低旋转段与用于臂长估计的足够旋转激励相结合）。

符号说明：在本文中，我们遵循以下符号约定：上标  $(\cdot)^\circ$  表示变量  $(\cdot)$  的真实值， $(\cdot)^\circ$  表示对  $(\cdot)$  的测量，而  $(\cdot)^\wedge$  表示对  $(\cdot)$  的估计。对于向量  $a \in \mathbb{R}^3$ ，设  $a^\wedge \in \mathbb{R}^{3 \times 3}$  为满足  $a \times b = a^\wedge b$  的斜对称矩阵，对于所有  $b \in \mathbb{R}^3$ ，其中  $\times$  是向量叉乘。

## 二、问题提出

### A. DVL 测量模型

当DVL安装在水下机器人上时，它会发射声波束并测量由机器人相对于其周围环境运动引起的多普勒频移，这使DVL能够在假设声速已知且恒定的情况下计算三维速度。

从数学上讲，DVL 测量是相对于其本地机体固定坐标系  $\{\mathcal{F}_d\}$  来解析的。为了分析 DVL 相对于基准坐标系  $\{\mathcal{F}_b\}$ （通常附着于其他传感器）的运动，记  $p_{o_d,b}$  为在坐标系  $\{\mathcal{F}_b\}$  中表示的  $\{\mathcal{F}_d\}$  的原点  $o_d$  的坐标。本质上，DVL 测量的是在其自身坐标系  $\{\mathcal{F}_d\}$  中表示的  $o_d$  的线速度。假设基准坐标系和 DVL 坐标系刚性连接，当基准以线速度  $v_b$  和角速度  $\omega_b$ （均以  $\{\mathcal{F}_b\}$  表示）运动时，DVL 速度  $v_d$  与  $(v_b, \omega_b)$  之间的关系是

$$v_d = {}^dR_b(v_b + \omega_b \times {}^d t_b), \quad (1)$$

其中  ${}^dR_b \in \text{SO}(3)$  是将坐标从  $\{\mathcal{F}_b\}$  映射到  $\{\mathcal{F}_d\}$  的旋转， ${}^d t_b \in \mathbb{R}^3$ ，等于  $p_{o_d,b}$ ，是从基座坐标系原点到 DVL 坐标系原点的力臂，用  $\{\mathcal{F}_b\}$  表示。

为了考虑声速不匹配（例如，由于温度和盐度）以及其他乘法尺度偏差，我们引入了一个正的缩放因子  $\kappa_d$ 。我们还

考虑一个加性测量噪声项。因此，实际的DVL测量  $\hat{v}_d$  可以建模为

$$\hat{v}_d = \kappa_d {}^dR_b(v_b + \omega_b \times {}^d t_b) + n_v, \quad (2)$$

$n_v \sim \mathcal{N}(0, Q_d)$  表示测量噪声，其协方差可以从 DVL 传感器获得。

### B. 基于高斯过程运动先验的统一时空校准问题

相对于固定世界坐标系的机器人运动由在基坐标系中表示的连续时间状态  $x(t)$  表示。我们将  $x(t)$  建模为高斯过程（GP），

$$x(t) \sim \mathcal{GP}(\mu(t), K(t, t')), \quad (3)$$

其中  $K(t, t') \triangleq \text{cov}(x(t), x(t'))$  是（矩阵值）GP 协方差核（具有选定的超参数）[17]。附加在底座的参考传感器套件提供了时间戳为  ${}^bT = \{{}^bT_j\}_{j=1}^M$  的测量  $\hat{\mathcal{M}}_b = \{\hat{m}_{b,j}\}_{j=1}^M$ ，测量模型为

$$\hat{m}_{b,j} = h(x({}^bT_j)) + n_{b,j}, \quad n_{b,j} \sim \mathcal{N}(0, R_{b,j}), \quad j = 1, \dots, M. \quad (4)$$

定义堆叠向量  $\hat{\mathcal{M}}_b \triangleq [\hat{m}_{b,1}^\top \cdots \hat{m}_{b,M}^\top]^\top$ ，并令  $H$  表示在 GP 回归使用的线性化点处评估的  $h(\cdot)$  的雅可比矩阵。令  $X_b \triangleq [x({}^bT_1)^\top \cdots x({}^bT_M)^\top]^\top$  为基准时间戳处的堆叠状态。使用  $\hat{\mathcal{M}}_b$  和 GP 先验，我们通过 GP 回归计算参考轨迹估计（及其不确定性），记为  $\hat{X}_b$ （和  $\Sigma_{X_b}$ ）；封闭形式的表达在第 III-B 节提供。

DVL 提供速度测量  $\hat{\mathcal{V}}_d = \{\hat{v}_{d,i}\}_{i=1}^N$ ，由其本地时钟在  ${}^dT = \{{}^dT_i\}_{i=1}^N$  打时间戳。基座时钟与 DVL 时钟之间的恒定时钟偏移由  ${}^b\delta_{T,d}$  建模，因此在  ${}^dT_i$  记录的 DVL 样本的物理时间对应于基座时间  ${}^dT_i + {}^b\delta_{T,d}$ 。感兴趣的校准变量被收集在

$$\Theta \triangleq \{{}^b\delta_{T,d}, {}^dR_b, {}^d t_b, \kappa_d\},$$

其中  ${}^dR_b \in \text{SO}(3)$ ， ${}^d t_b \in \mathbb{R}^3$  为外参， $\kappa_d > 0$  为 DVL 速度尺度因子。对于候选  ${}^b\delta_{T,d}$ ，通过  ${}^bT_{d,i}({}^b\delta_{T,d}) \triangleq {}^dT_i + {}^b\delta_{T,d}$  定义基准时钟下对齐的 DVL 时间戳，并让  $X_d({}^b\delta_{T,d}) \triangleq [x({}^bT_{d,1}({}^b\delta_{T,d}))^\top \cdots x({}^bT_{d,N}({}^b\delta_{T,d}))^\top]^\top$  表示在这些对齐时刻查询到的状态。已知  $\hat{X}_b$ ，查询到的均值  $\hat{X}_d({}^b\delta_{T,d})$  可通过 GP 插值获得（作为  ${}^b\delta_{T,d}$  的显函数），相应的查询协方差  $\Sigma_{X_d}({}^b\delta_{T,d})$  可作为 GP 回归和插值的副产品，根据 (8) 和 (9) 计算。

在对齐时间  ${}^bT_{d,i}({}^b\delta_{T,d})$ ，我们期望 DVL 测量为  $\kappa_d {}^dR_b(\hat{v}_{b,i}({}^b\delta_{T,d}) + \hat{\omega}_{b,i}({}^b\delta_{T,d}) \times {}^d t_b)$ ，其中  $\hat{v}_{b,i}({}^b\delta_{T,d})$  和  $\hat{\omega}_{b,i}({}^b\delta_{T,d})$  是从查询状态中提取的线性和角速度分量

$\hat{X}_{d,i}(^b\delta_{T,d})$ . We then define the DVL least-squares cost

$$J_{d,M}(\hat{X}_d(^b\delta_{T,d}), ^dR_b, ^d\mathbf{t}_b, \kappa_d) \triangleq \sum_{i=1}^N \left\| \hat{v}_{d,i} - \kappa_d ^dR_b (\hat{v}_{b,i}(^b\delta_{T,d}) + \hat{\omega}_{b,i}(^b\delta_{T,d}) \times ^d\mathbf{t}_b) \right\|_{Q_d}^2. \quad (5)$$

We pose spatiotemporal calibration as the following constrained nonlinear least-squares problem:

$$\begin{aligned} \min_{^b\delta_{T,d}, ^dR_b, ^d\mathbf{t}_b, \kappa_d} \quad & J_{d,M}(\hat{X}_d(^b\delta_{T,d}), ^dR_b, ^d\mathbf{t}_b, \kappa_d) \\ \text{s.t.} \quad & ^b\delta_{T,d} \in \mathbb{R}, ^dR_b \in \text{SO}(3), ^d\mathbf{t}_b \in \mathbb{R}^3, \kappa_d > 0. \end{aligned} \quad (6)$$

Problem (6) is an error-in-variables (EIV) problem [18]–[20], because the queried trajectory  $\hat{X}_d(^b\delta_{T,d})$  is itself estimated from noisy reference measurements but is independent of the DVL noise. The problem is also highly nonconvex owing to the coupled spatiotemporal parameters and time-shifted GP queries. We therefore develop a sequential estimator of  $\Theta$  to initialize the subsequent local nonlinear least-squares refinement.

### III. METHODOLOGY

UIC consists of sequential initialization followed by local iterative refinement. The initializer estimates all components of  $\Theta$  without manual or near-true guesses. The reference trajectory  $x(t)$  is inferred from the base measurements under the GP prior and is not jointly optimized with  $\Theta$ , because a flexible trajectory could absorb spatiotemporal miscalibration and confound parameter estimation. Under the idealized EIV assumptions, the initializer is designed to enter a neighborhood of the true parameters, after which refinement further decreases the residual.

#### A. Sequential Initialization Strategy

Our initialization procedure separates reference motion estimation from the calibration parameters  $^dR_b$ ,  $^d\mathbf{t}_b$ ,  $\kappa_d$ , and  $^b\delta_{T,d}$ . Concretely, UIC first computes a reference trajectory estimate  $\hat{X}_b$  and its uncertainty  $\Sigma_{X_b}$  from the base measurements  $\dot{M}_b$  under the GP motion prior. Then, for any candidate clock offset  $^b\delta_{T,d}$ , the time-shifted motion states at the aligned DVL timestamps are obtained by GP interpolation, yielding  $\hat{X}_d(^b\delta_{T,d})$  and  $\Sigma_{X_d}(^b\delta_{T,d})$ . This stage does not use DVL measurements, so that the reference motion estimate remains independent of the unknown calibration parameters and admits explicit GP-based interpolation. According to [21], the GP regression/interpolation stage can be implemented efficiently and provides both a reference motion estimate and an uncertainty characterization, which are useful for subsequent calibration. The large-sample analysis is derived under idealized modeling assumptions (e.g., correct GP regression/interpolation statistics and sufficiently excited motion), and is intended to justify the design of the initialization estimators rather than to serve as finite-sample performance guarantees.

Directly optimizing Problem (6) from random initialization offers no strong convergence guarantees. Poor initial guesses often lead to divergence or convergence to suboptimal local minima. In our experiments, naive initialization frequently resulted in non-convergence. The initialization procedures are presented in the next two subsections.

#### B. Initialization of Motion State $X$

By GP regression conditioned on the base measurements  $\dot{M}_b$  and the GP prior, we obtain an estimate  $\hat{X}_b$  of the base states  $X_b$  in closed form. Then, for any candidate clock offset  $^b\delta_{T,d}$ , the queried states at the aligned DVL timestamps are computed as  $\hat{X}_d(^b\delta_{T,d})$  via standard GP interpolation. Let  $^bT_d(^b\delta_{T,d}) \triangleq \{^bT_{d,i}(^b\delta_{T,d})\}_{i=1}^N$ , where  $^bT_{d,i}(^b\delta_{T,d}) \triangleq ^dT_i + ^b\delta_{T,d}$ . By abuse of notation,  $K(T_1, T_2)$  denotes the block (cross-)covariance matrix assembled from the GP kernel  $K(t, t')$ . Define  $K_1 \triangleq K(^bT, ^bT)$ ,  $K_2(^b\delta_{T,d}) \triangleq K(^bT, ^bT_d(^b\delta_{T,d}))$ ,  $K_3(^b\delta_{T,d}) \triangleq K(^bT_d(^b\delta_{T,d}), ^bT)$ , and  $K_4(^b\delta_{T,d}) \triangleq K(^bT_d(^b\delta_{T,d}), ^bT_d(^b\delta_{T,d}))$ . Similarly  $\mu = \begin{bmatrix} \mu_1 \\ \mu_2(^b\delta_{T,d}) \end{bmatrix}$  which  $\mu_1$  denote the GP prior mean at  $^bT$  and  $\mu_2$  denote the GP prior mean at  $^bT_d(^b\delta_{T,d})$ . According to the linear prediction equations [21, (50)],  $\hat{X}_b$  is estimated via GP regression as

$$\hat{X}_b = \mu_1 + K_1 H^\top (R + H K_1 H^\top)^{-1} (M_b - H \mu_1). \quad (7)$$

Then  $\hat{X}_d$  is computed as a linear combination of  $\hat{X}_b$ , with weights determined explicitly by  $^bT$ ,  $^dT$ , and  $^b\delta_{T,d}$ , i.e.,

$$\hat{X}_d(^b\delta_{T,d}) = \mu_2(^b\delta_{T,d}) - K_3(^b\delta_{T,d}) K_1^{-1} (\mu_1 - \hat{X}_b). \quad (8)$$

This makes  $\hat{X}_d(^b\delta_{T,d})$  an explicit function of the time offset through the GP interpolation weights. In practice, the GP regression in (7) and the query in (8) can be implemented either in closed form for moderate problem sizes, or via sparse/Markov representations when the chosen kernel admits a state-space form; both filtering/smoothing and factor-graph solvers are applicable. In addition, the posterior uncertainty can be computed as a byproduct of GP regression/interpolation [17], given by

$$\begin{aligned} \Sigma_{X_b} &= K_1 - K_1 H^\top (R + H K_1 H^\top)^{-1} H K_1, \\ \Sigma_{X_d}(^b\delta_{T,d}) &= K_4(^b\delta_{T,d}) - \\ &\quad K_3(^b\delta_{T,d}) K_1^{-1} (K_1 - \Sigma_{X_b}) K_1^{-1} K_2(^b\delta_{T,d}). \end{aligned} \quad (9)$$

#### C. Initialization of Calibration Variables

We next initialize the calibration variables using all DVL measurements together with the GP estimate  $\hat{X}_d(^b\delta_{T,d})$ . We decouple the calibration of the extrinsic parameters  $^dR_b$  and  $^d\mathbf{t}_b$  from that of the temporal offset and scale parameters  $^b\delta_{T,d}$  and  $\kappa_d$  by leveraging specialized robot motion patterns. As will be shown shortly, the resulting estimators can be computed efficiently—via line



$\hat{X}_{d,i}(^b\delta_{T,d})$ 然后我们定义 DVL 最小二乘代价

$$J_{d,M}(\hat{X}_d(^b\delta_{T,d}), {}^dR_b, {}^d\mathbf{t}_b, \kappa_d) \triangleq \sum_{i=1}^N \left\| \hat{v}_{d,i} - \kappa_d {}^dR_b(\hat{v}_{b,i}(^b\delta_{T,d}) + \hat{\omega}_{b,i}(^b\delta_{T,d}) \times {}^d\mathbf{t}_b) \right\|_{Q_d}^2. \quad (5)$$

我们将时空校准表示为以下受约束的非线性最小二乘问题：

$$\begin{aligned} \min_{^b\delta_{T,d}, {}^dR_b, {}^d\mathbf{t}_b, \kappa_d} \quad & J_{d,M}(\hat{X}_d(^b\delta_{T,d}), {}^dR_b, {}^d\mathbf{t}_b, \kappa_d) \\ \text{s.t.} \quad & {}^b\delta_{T,d} \in \mathbb{R}, {}^dR_b \in \text{SO}(3), {}^d\mathbf{t}_b \in \mathbb{R}^3, \kappa_d > 0. \end{aligned} \quad (6)$$

问题 (6) 是一个误差-变量 (EIV) 问题 [18]–[20]，因为被查询的轨迹  $\hat{X}_d(^b\delta_{T,d})$  本身是从带噪声的参考测量中估计出来的，但与 DVL 噪声无关。由于耦合的时空参数和时间偏移的 GP 查询，该问题也是高度非凸的。因此，我们开发了一个  $\Theta$  的顺序估计器，以初始化随后进行的局部非线性最小二乘优化。

### 三、方法论

UIC 由顺序初始化后跟局部迭代优化组成。初始化器在没有人工或接近真实的猜测情况下估计  $\Theta$  的所有组成部分。参考轨迹  $x(t)$  是在 GP 先验下从基础测量推断出来的，并且不会与  $\Theta$  一起联合优化，因为灵活的轨迹可能会吸收时空误差校准并混淆参数估计。在理想化的 EIV 假设下，初始化器旨在进入真实参数的邻域，然后进一步的优化可以降低残差。

#### A. 顺序初始化策略

我们的初始化过程将参考运动估计与校准参数  ${}^dR_b$ 、 ${}^d\mathbf{t}_b$ 、 $\kappa_d$  和  ${}^b\delta_{T,d}$  分开。具体而言，UIC 首先根据 GP 运动先验，从基础测量  $\dot{M}_b$  中计算参考轨迹估计  $\hat{X}_b$  及其不确定性  $\Sigma_{X_b}$ 。然后，对于任意候选时钟偏移  ${}^b\delta_{T,d}$ ，通过 GP 插值获得在对齐的 DVL 时间戳下的时间偏移运动状态，从而得到  $\hat{X}_d(^b\delta_{T,d})$  和  $\Sigma_{X_d}(^b\delta_{T,d})$ 。该阶段不使用 DVL 测量，因此参考运动估计仍然独立于未知校准参数，并且允许基于 GP 的显式插值。根据 [21]，GP 回归/插值阶段可以高效实现，并提供参考运动估计及不确定性表征，这对于后续校准非常有用。在理想化建模假设（例如，正确的 GP 回归/插值统计和足够的样本量）下派生了大样本分析。

直接从随机初始化优化问题 (6) 并不能提供强有力的收敛保证。较差的初始猜测常常导致发散或收敛到次优的局部极小值。在我们的实验中，简单的初始化经常导致不收敛。初始化步骤将在接下来的两个小节中介绍。

#### B. 运动状态的初始化 $X$

通过以基础测量值  $\dot{M}_b$  和 GP 先验为条件的 GP 回归，我们以闭合形式获得基础状态  $X_b$  的估计  $\hat{X}_b$ 。然后，对于任意候选时钟偏移  ${}^b\delta_{T,d}$ ，在对齐的 DVL 时间戳下查询的状态可通过标准 GP 插值计算为  $\hat{X}_d(^b\delta_{T,d})$ 。设  ${}^bT_d(^b\delta_{T,d}) \triangleq \{{}^bT_{d,i}(^b\delta_{T,d})\}_{i=1}^N$ ，其中  ${}^bT_{d,i}(^b\delta_{T,d}) \triangleq {}^dT_i + {}^b\delta_{T,d}$ 。滥用符号， $K(T_1, T_2)$  表示由 GP 核  $K(t, t')$  组装的块（交叉）协方差矩阵。定义  $K_1 \triangleq K(^bT, ^bT)$ 、 $K_2(^b\delta_{T,d}) \triangleq K(^bT, {}^bT_d(^b\delta_{T,d}))$ 、 $K_3(^b\delta_{T,d}) \triangleq K(^bT_d(^b\delta_{T,d}), ^bT)$  和  $K_4(^b\delta_{T,d}) \triangleq K(^bT_d(^b\delta_{T,d}), {}^bT_d(^b\delta_{T,d}))$ 。同样， $\mu = \begin{bmatrix} \mu_1 \\ \mu_2 \end{bmatrix}$  其  $\mu_1$  表示  ${}^bT$  处的 GP 先验均值，而  $\mu_2$  表示  ${}^bT_d(^b\delta_{T,d})$  处的 GP 先验均值。根据线性预测方程 [21, (5 0)]， $\hat{X}_b$  通过 GP 回归估计为

$$\hat{X}_b = \mu_1 + K_1 H^\top (R + H K_1 H^\top)^{-1} (M_b - H \mu_1). \quad (7)$$

然后  $\hat{X}_d$  被计算为  $\hat{X}_b$  的线性组合，其权重由  ${}^bT$ 、 ${}^dT$  和  ${}^b\delta_{T,d}$  明确确定，即，

$$\hat{X}_d(^b\delta_{T,d}) = \mu_2(^b\delta_{T,d}) - K_3(^b\delta_{T,d}) K_1^{-1} (\mu_1 - \hat{X}_b). \quad (8)$$

这使得  $\hat{X}_d(^b\delta_{T,d})$  通过 GP 插值权重成为时间偏移的显式函数。在实践中，(7) 中的 GP 回归和 (8) 中的查询可以针对适中的问题规模以闭式形式实现，或在所选核函数允许状态空间形式的情况下，通过稀疏/马尔可夫表示来实现；滤波/平滑和因子图求解器均适用。此外，后验不确定性可以作为 GP 回归/插值 [17] 的副产品计算，表示为

$$\begin{aligned} \Sigma_{X_b} &= K_1 - K_1 H^\top (R + H K_1 H^\top)^{-1} H K_1, \\ \Sigma_{X_d}(^b\delta_{T,d}) &= K_4(^b\delta_{T,d}) - \\ &\quad K_3(^b\delta_{T,d}) K_1^{-1} (K_1 - \Sigma_{X_b}) K_1^{-1} K_2(^b\delta_{T,d}). \end{aligned} \quad (9)$$

#### C. 校准变量的初始化

接下来，我们使用所有 DVL 测量数据以及 GP 估计  $\hat{X}_d(^b\delta_{T,d})$  来初始化校准变量。我们通过利用专业化的机器人运动模式，将外参数  ${}^dR_b$  和  ${}^d\mathbf{t}_b$  的校准与时间偏移和尺度参数  ${}^b\delta_{T,d}$  和  $\kappa_d$  的校准解耦。如将很快展示的那样，得到的估计器可以通过线性方法高效计算。

search for the temporal/scale parameters, and a closed-form bias-corrected estimator for the spatial extrinsics.

1) Clock offset and scale initialization: In principle, we estimate the clock offset  ${}^b\delta_{T,d}$  by maximizing the alignment between the DVL measurements and the predicted DVL output based on  $\hat{X}_d({}^b\delta_{T,d})$ . Specifically, the predicted DVL velocity is given by  $\kappa_d {}^dR_b (\hat{v}_{b,i}({}^b\delta_{T,d}) + \hat{\omega}_{b,i}({}^b\delta_{T,d}) \times {}^d\mathbf{t}_b)$  according to (2). To reduce the influence of the spatial extrinsics in this stage, we select a subset of measurements  $\mathcal{D}_o$  with small angular velocity, i.e.,  $\omega_{b,i} \approx 0$ . Under this operating regime, the lever-arm term  $\hat{\omega}_{b,i}({}^b\delta_{T,d}) \times {}^d\mathbf{t}_b$  becomes negligible. Furthermore, taking the Euclidean norm removes the effect of  ${}^dR_b$  due to rotation invariance. Combining these yields, for  $i \in \mathcal{D}_o$ ,

$$\kappa_d^2 \|\hat{v}_{b,i}({}^b\delta_{T,d})\|^2 \approx \|v_{d,i}^o\|^2. \quad (10)$$

Centering the noise terms leads to the following two-dimensional search problem:

$$\min_{{}^b\delta_{T,d} \kappa_d} \sum_{i \in \mathcal{D}_o} \left( \frac{\|\hat{v}_{d,i}\|^2 - \text{tr}(Q_d)}{\kappa_d^2} \|\hat{v}_{b,i}({}^b\delta_{T,d})\|^2 + \text{tr}(\Sigma_{v_b,i}({}^b\delta_{T,d})) \right), \quad (11)$$

where  $\Sigma_{v_b,i}({}^b\delta_{T,d})$  denotes the  $3 \times 3$  marginal covariance of the interpolated linear velocity  $\hat{v}_{b,i}({}^b\delta_{T,d})$ , extracted from  $\Sigma_{X_d}({}^b\delta_{T,d})$ . The problem can be efficiently solved by performing a grid search over  ${}^b\delta_{T,d}$  while varying  $\kappa_d$  over a small grid around 1. Let  ${}^b\hat{\delta}_{T,d}$  and  $\hat{\kappa}_d$  be any global minimizer. The theorem characterizes large-sample behavior under the idealized regime  $\omega_{b,i} = 0$  for  $i \in \mathcal{D}_o$ .

Remark 1: Under idealized conditions (e.g., sufficiently small angular rates on the selected subset, correct second-moment models for GP interpolation errors, and identifiable speed excitation), the minimizer of (11) is a consistent estimator of  $({}^b\delta_{T,d}, \kappa_d)$  as the number of selected samples increases. This remark follows standard EIV-estimation and is included to motivate the estimator structure rather than to provide a finite-sample guarantee.

2) Bias-corrected closed-form extrinsics initialization: Stacking all DVL measurements as  $\hat{V}_d = [\hat{v}_{d,1} \cdots \hat{v}_{d,N}]^\top$ , we arrive at the stacked version of (2):

$$\hat{V}_d = \kappa_d V_b^o {}^dR_b^\top - \kappa_d \Omega_b^o ({}^d\mathbf{t}_b)^\top {}^dR_b^\top + N_v, \quad (12)$$

where  $V_b^o = [v_{b,1}^o \cdots v_{b,N}^o]^\top$ ,  $\Omega_b^o = [\omega_{b,1}^o \cdots \omega_{b,N}^o]^\top$  and  $N_v = [n_{v,1} \cdots n_{v,N}]^\top$ . Define

$$Z_1 = \kappa_d {}^dR_b^\top, \quad Z_2 = \kappa_d ({}^d\mathbf{t}_b)^\top {}^dR_b^\top, \quad Z \triangleq \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} \in \mathbb{R}^{6 \times 3},$$

and define  $H^o \triangleq [V_b^o \quad -\Omega_b^o] \in \mathbb{R}^{N \times 6}$ . Then (12) can be written compactly as  $\hat{V}_d = H^o Z + N_v$ . Omitting the SO(3) constraint of  ${}^dR_b$ , we estimate  $Z$  by solving

$$\argmin_Z \|\hat{V}_d - H^o Z\|_F^2,$$

which admits the closed-form solution

$$\hat{Z} = \left( \frac{H^{o\top} H^o}{N} \right)^{-1} \left( \frac{H^{o\top} \hat{V}_d}{N} \right). \quad (13)$$

We note that  $\kappa_d$  is estimated again in the above problem. This is a deliberate technical choice: the earlier estimation used only a selected subset  $\mathcal{D}_o$  of DVL measurements with  $\omega_{b,i} \approx 0$ , which represents a relatively limited volume of data. In contrast, the stacked formulation (12) uses the entire measurement set. This yields a reliable estimate of  $\kappa_d$  in the sense that it exhibits less variance around the true value.

The exact values of  $v_{b,i}^o$  and  $\omega_{b,i}^o$  are not directly measurable, therefore their estimates  $\hat{v}_{b,i}$  and  $\hat{\omega}_{b,i}$  from  $\hat{X}_b$  are substituted into (13). Accordingly,  $\hat{Z}$  should be corrected to account for this substitution in the EIV setting. The resulting bias-corrected estimate is

$$\hat{Z}^c = \left( \frac{\hat{H}^\top \hat{H} - G}{N} \right)^{-1} \left( \frac{\hat{H}^\top \hat{V}_d}{N} \right), \quad (14)$$

where  $\hat{H} = [\hat{V}_b - \hat{\Omega}_b]$  with  $\hat{V}_b = [\hat{v}_{b,1}({}^b\hat{\delta}_{T,d}) \cdots \hat{v}_{b,N}({}^b\hat{\delta}_{T,d})]$  and  $\hat{\Omega}_b = [\hat{\omega}_{b,1}({}^b\hat{\delta}_{T,d}) \cdots \hat{\omega}_{b,N}({}^b\hat{\delta}_{T,d})]$ , and  $G \in \mathbb{R}^{6 \times 6}$  is the covariance-induced correction term. Specifically, letting  $\Sigma_{v_b,i}({}^b\hat{\delta}_{T,d})$  and  $\Sigma_{\omega_b,i}({}^b\hat{\delta}_{T,d})$  denote the  $3 \times 3$  marginal covariances of the interpolated linear and angular velocities (extracted from  $\Sigma_{X_d}({}^b\hat{\delta}_{T,d})$  in (9)), we define

$$G \triangleq \sum_{i=1}^N \text{blkdiag}(\Sigma_{v_b,i}({}^b\hat{\delta}_{T,d}), \Sigma_{\omega_b,i}({}^b\hat{\delta}_{T,d})).$$

We further reconstruct  $\hat{\kappa}_d^c$  from  $\hat{Z}^c$  as

$$\hat{\kappa}_d^c = |\det(\hat{Z}_1^c)|^{1/3}, \quad (15)$$

which follows from  $\det({}^d\hat{R}_b^o) = 1$ . Performing singular value decomposition (SVD) on  $(\hat{Z}_1^c / \hat{\kappa}_d^c)^\top$ , we write  $(\hat{Z}_1^c / \hat{\kappa}_d^c)^\top = U_R \Sigma_R V_R^\top$ , and set

$${}^d\hat{R}_b^c = U_R \text{diag}([1, 1, \det(U_R V_R^\top)]) V_R^\top. \quad (16)$$

We then recover an estimate  ${}^d\hat{\mathbf{t}}_b^{c\wedge}$  from  $\hat{Z}_2^c$  as

$${}^d\hat{\mathbf{t}}_b^{c\wedge} = \left( -\hat{Z}_2^c {}^d\hat{R}_b^c / \hat{\kappa}_d^c - (-\hat{Z}_2^c {}^d\hat{R}_b^c / \hat{\kappa}_d^c)^\top \right) / 2. \quad (17)$$

The following result characterizes the large-sample behavior of the above estimates under standard EIV conditions.

Remark 2: When the motion is sufficiently exciting so that the normal matrix is nonsingular, and when the GP-interpolated velocity errors satisfy standard regularity conditions, the structured estimator recovered from (14)–(17) is consistent under standard EIV arguments. This remark is primarily included to motivate the bias-correction structure; finite-sample behavior is assessed experimentally.

搜索时间/尺度参数，以及用于空间外参数的闭式偏差校正估计量。

1) 时钟偏移和尺度初始化：原则上，我们通过最大化 DVL 测量值与基于  $\hat{X}_d(b\delta_{T,d})$  预测的 DVL 输出之间的对齐来估计时钟偏移  $b\delta_{T,d}$ 。具体来说，根据公式 (2)，预测的 DVL 速度由  $\kappa_d {}^dR_b(\hat{v}_{b,i}(b\delta_{T,d}) + \hat{\omega}_{b,i}(b\delta_{T,d}) \times \hat{t}_b)$  给出。为了在此阶段减少空间外参的影响，我们选择角速度较小的测量子集  $\mathcal{D}_o$ ，即  $\omega_{b,i} \approx 0$ 。在这种操作状态下，杆臂项  $\hat{\omega}_{b,i}(b\delta_{T,d}) \times \hat{t}_b$  可忽略不计。此外，通过取欧几里得范数可以消除由于旋转不变性带来的  ${}^dR_b$  影响。将这些结合起来，对于  $i \in \mathcal{D}_o$ ，

$$\kappa_d^2 \|\hat{v}_{b,i}(b\delta_{T,d})\|^2 \approx \|v_{d,i}^o\|^2. \quad (10)$$

对噪声项进行中心化会导致以下二维搜索问题：

$$\min_{b\delta_{T,d}} \sum_{i \in \mathcal{D}_o} \left( \frac{\|\hat{v}_{d,i}\|^2 - \text{tr}(Q_d)}{\kappa_d^2} \|\hat{v}_{b,i}(b\delta_{T,d})\|^2 + \text{tr}(\Sigma_{v_{b,i}}(b\delta_{T,d})) \right)^2, \quad (11)$$

其中  $\Sigma_{v_{b,i}}(b\delta_{T,d})$  表示插值线速度  $\hat{v}_{b,i}(b\delta_{T,d})$  的  $3 \times 3$  边际协方差，该协方差是从  $\Sigma_{X_d}(b\delta_{T,d})$  提取的。这个问题可以通过在  $b\delta_{T,d}$  上执行网格搜索，同时在以 1 为中心的小网格上变化  $\kappa_d$  来高效解决。设  $b\hat{\delta}_{T,d}$  和  $\hat{\kappa}_d$  为任意全局最小值。该定理刻画了理想化情形  $\omega_{b,i} = 0$  下  $i \in \mathcal{D}_o$  的大样本行为。

备注1：在理想条件下（例如，在选定子集上的角速度足够小、用于GP插值误差的二阶矩模型正确，以及速度激励可识别），随着所选样本数量的增加，(11) 的最小化器是  $(b\delta_{T,d}, \kappa_d)$  的一致估计量。此备注遵循标准的EIV估计方法，包含此内容的目的是为了说明估计器结构，而不是提供有限样本的保证。

2) 偏差校正的闭式外参初始化：将所有 DVL 测量值堆叠为  $\hat{V}_d = [\hat{v}_{d,1} \cdots \hat{v}_{d,N}]^\top$ ，我们得到公式 (2) 的堆叠版本：

$$\hat{V}_d = \kappa_d V_b^o {}^dR_b^\top - \kappa_d \Omega_b^o (t_b^\wedge)^\top {}^dR_b^\top + N_v, \quad (12)$$

哪里  $V_b^o = [v_{b,1}^o \cdots v_{b,N}^o]^\top$ ， $\Omega_b^o = [\omega_{b,1}^o \cdots \omega_{b,N}^o]^\top$  和  $N_v = [n_{v,1} \cdots n_{v,N}]^\top$ 。定义

$$Z_1 = \kappa_d {}^dR_b^\top, \quad Z_2 = \kappa_d (t_b^\wedge)^\top {}^dR_b^\top, \quad Z \triangleq \begin{bmatrix} Z_1 \\ Z_2 \end{bmatrix} \in \mathbb{R}^{6 \times 3},$$

并定义  $H^o \triangleq [V_b^o \quad -\Omega_b^o] \in \mathbb{R}^{N \times 6}$ 。然后 (12) 可以简洁地写为  $\hat{V}_d = H^o Z + N_v$ 。省略  ${}^dR_b$  的 SO(3) 约束，我们通过求解来估计  $Z$

$$\underset{Z}{\operatorname{argmin}} \quad \|\hat{V}_d - H^o Z\|_F^2,$$

承认封闭形式解

$$\hat{Z} = \left( \frac{H^{o\top} H^o}{N} \right)^{-1} \left( \frac{H^{o\top} \hat{V}_d}{N} \right). \quad (13)$$

我们注意到，在上述问题中再次对  $\kappa_d$  进行了估算。这是一个有意的技术选择：之前的估算仅使用了 DVL 测量的一个选定子集  $\mathcal{D}_o$ ，该子集的  $\omega_{b,i} \approx 0$ ，这代表了相对有限的数量。相比之下，堆叠公式 (12) 使用了整个测量集合。这在某种意义上提供了对  $\kappa_d$  的可靠估算，因为它在真值附近表现出较小的方差。

$v_{b,i}^o$  和  $\omega_{b,i}^o$  的精确数值无法直接测量，因此将其从  $\hat{X}_b$  得到的估算值  $\hat{v}_{b,i}$  和  $\hat{\omega}_{b,i}$  代入 (13) 中。相应地， $\hat{Z}$  应在 E IV 设置中进行修正，以考虑这一替代。由此得到的偏差修正估计值为

$$\hat{Z}^c = \left( \frac{\hat{H}^\top \hat{H} - G}{N} \right)^{-1} \left( \frac{\hat{H}^\top \hat{V}_d}{N} \right), \quad (14)$$

其中  $\hat{H} = [\hat{V}_b - \hat{\Omega}_b]$  与  $\hat{V}_b = [\hat{v}_{b,1}(b\hat{\delta}_{T,d}) \cdots \hat{v}_{b,N}(b\hat{\delta}_{T,d})]$  和  $\hat{\Omega}_b = [\hat{\omega}_{b,1}(b\hat{\delta}_{T,d}) \cdots \hat{\omega}_{b,N}(b\hat{\delta}_{T,d})]$ ，并且  $G \in \mathbb{R}^{6 \times 6}$  是协方差引起的修正项。具体而言，令  $\Sigma_{v_{b,i}}(b\hat{\delta}_{T,d})$  和  $\Sigma_{\omega_{b,i}}(b\hat{\delta}_{T,d})$  表示插值线速度和角速度（从公式 (9) 中的  $\Sigma_{X_d}(b\hat{\delta}_{T,d})$  提取）的  $3 \times 3$  边际协方差，我们定义

$$G \triangleq \sum_{i=1}^N \text{blkdiag}(\Sigma_{v_{b,i}}(b\hat{\delta}_{T,d}), \Sigma_{\omega_{b,i}}(b\hat{\delta}_{T,d})).$$

我们进一步从  $\hat{Z}^c$  重建  $\hat{\kappa}_d^c$  为

$$\hat{\kappa}_d^c = |\det(\hat{Z}_1^c)|^{1/3}, \quad (15)$$

这由  $\det({}^d\hat{R}_b^o) = 1$  推出。1. 对  $(\hat{Z}_1^c/\hat{\kappa}_d^c)^\top$  进行奇异值分解 (SVD)，我们写作  $(\hat{Z}_1^c/\hat{\kappa}_d^c)^\top = U_R \Sigma_R V_R^\top$ ，并设置

$${}^d\hat{R}_b^c = U_R \text{diag}([1, 1, \det(U_R V_R^\top)]) V_R^\top. \quad (16)$$

然后我们从  $\hat{Z}_2^c$  恢复  $t_b^{c\wedge}$  的估计值为

$$t_b^{c\wedge} = \left( -\hat{Z}_2^c {}^d\hat{R}_b^c / \hat{\kappa}_d^c - (-\hat{Z}_2^c {}^d\hat{R}_b^c / \hat{\kappa}_d^c)^\top \right) / 2. \quad (17)$$

以下结果描述了在标准误差-变量(EIV)条件下上述估计的大样本行为。

备注2：当运动足够激励以至于法矩阵是非奇异的，并且当 GP 插值的速度误差满足标准正则条件时，从 (14)-(17) 中恢复的结构化估计量在标准 EIV 论证下是一致的。该备注主要是为了说明偏差校正结构的动机；有限样本行为通过实验进行评估。



## D. Iterative Refinement

Starting from the sequential initialization  $\hat{\Theta}$ , we refine  $\Theta$  by local iterative optimization of Problem (6). The rotation is updated on  $\text{SO}(3)$  via an exponential-map perturbation, and the DVL scale is re-parameterized to enforce  $\kappa_d > 0$ :

$${}^dR_b \leftarrow \exp(\phi^\wedge) {}^dR_b, \quad \phi \in \mathbb{R}^3, \quad (18)$$

$$\kappa_d = \exp(s_d), \quad s_d \in \mathbb{R}. \quad (19)$$

We apply numerical gradient descent to refine the parameters.

## E. Trajectory Design

The calibration accuracy of  $\Theta$  depends critically on motion excitation, which can be characterized via a Fisher information matrix (FIM) induced by the DVL velocity residual model in (5). Importantly, the predicted body-frame motion  $\hat{v}_b(\cdot)$  and  $\hat{\omega}_b(\cdot)$  is obtained from the continuous-time GP and therefore carries uncertainty. To reflect this, we use an effective residual covariance  $Q_e$ , which captures not only the intrinsic DVL measurement noise, but also the uncertainty induced by the estimated motion signals  $\hat{v}_b, \hat{\omega}_b$ .

Define the predicted DVL velocity at aligned time  ${}^bT_{d,i}({}^b\delta_{T,d}) = {}^dT_i + {}^b\delta_{T,d}$  as

$$\bar{v}_{d,i}(\Theta) = \kappa_d {}^dR_b \left( \hat{v}_{b,i}({}^b\delta_{T,d}) + \hat{\omega}_{b,i}({}^b\delta_{T,d}) \times {}^d\mathbf{t}_b \right).$$

Denote  $r_i(\Theta) \triangleq \hat{v}_{d,i} - \bar{v}_{d,i}(\Theta)$  and  $y_i({}^b\delta_{T,d}) \triangleq \hat{v}_{b,i}({}^b\delta_{T,d}) + \hat{\omega}_{b,i}({}^b\delta_{T,d}) \times {}^d\mathbf{t}_b$ . Using the unconstrained scale parameterization  $\kappa_d = \exp(s_d)$  and a left perturbation  ${}^dR_b \leftarrow \exp(\phi^\wedge) {}^dR_b$ , the first-order residual Jacobians satisfy

$$\begin{aligned} J_{s,i} &\triangleq \frac{\partial r_i}{\partial s_d} = -\kappa_d {}^dR_b y_i({}^b\delta_{T,d}), \\ J_{\phi,i} &\triangleq \frac{\partial r_i}{\partial \phi} = \kappa_d ({}^dR_b y_i({}^b\delta_{T,d}))^\wedge, \\ J_{t,i} &\triangleq \frac{\partial r_i}{\partial ({}^d\mathbf{t}_b)} = -\kappa_d {}^dR_b (\hat{\omega}_{b,i}({}^b\delta_{T,d}))^\wedge, \\ J_{\delta,i} &\triangleq \frac{\partial r_i}{\partial {}^b\delta_{T,d}} = -\kappa_d {}^dR_b \left( \dot{\hat{v}}_{b,i}({}^b\delta_{T,d}) + \dot{\hat{\omega}}_{b,i}({}^b\delta_{T,d}) \times {}^d\mathbf{t}_b \right), \end{aligned}$$

where  $(\dot{\cdot})$  denotes a time derivative evaluated at the aligned instant, which is always available analytically under a continuous-time GP model.

The FIM can be written in the stacked form

$$\mathcal{I}(\Theta) = J^\top Q_e^{-1} J, \quad J \triangleq \begin{bmatrix} J_1 \\ \vdots \\ J_N \end{bmatrix}, \quad J_i \triangleq [J_{\delta,i} \quad J_{s,i} \quad J_{\phi,i} \quad J_{t,i}].$$

Here  $Q_e$  is the effective covariance of the residuals, which aggregates the DVL measurement uncertainty and the additional uncertainty caused by motion estimates.

From the FIM, the lever arm  ${}^d\mathbf{t}_b$  is excited only by rotation since  $J_{t,i} \propto (\hat{\omega}_{b,i}({}^b\delta_{T,d}))^\wedge$ ; thus  $\hat{\omega}_{b,i}({}^b\delta_{T,d}) \approx 0$  makes  ${}^d\mathbf{t}_b$  unobservable, and near single-axis rotation yields an ill-conditioned  $\mathcal{I}_{tt}$  with the component of  ${}^d\mathbf{t}_b$

---

## Algorithm 1 UIC: Sequential Initialization + Gradient-Descent Refinement

---

Input: time-stamped DVL measurements  $\{(\hat{v}_{d,i}, T_{d,i})\}_{i=1}^N$ ; time-stamped base measurements  $\{(\hat{m}_{b,j}, T_{b,j})\}_{j=1}^M$ ; GP kernel  $K(t, t')$  in (3); angular-rate threshold  $\omega_{\text{th}}$  for selecting  $\mathcal{D}_o$

Output:  ${}^b\hat{\delta}_{T,d}$ ,  $\hat{\kappa}_d$ ,  ${}^d\hat{R}_b$ ,  ${}^d\hat{\mathbf{t}}_b$

- 1: GP reference trajectory: compute  $\hat{X}_b(\cdot)$  with covariances from  $\{\hat{m}_{b,j}\}$  and  $K$  via (7)–(9).
  - 2: Temporal/scale pre-initialization on  $\mathcal{D}_o$ : solve (11) by a grid search over  ${}^b\delta_{T,d}$  and  $\kappa_d$  around 1 only on the low-angular-velocity index set.
  - 3: Bias-corrected stacked extrinsics initialization: using  ${}^b\hat{\delta}_{T,d,\text{init}}$ , construct  $\hat{H}$  and  $G$  as in (14), and compute  $\hat{Z}^c$  using all  $\hat{v}_{d,i}$ .
  - 4: Recover scale and extrinsics: compute  $\hat{\kappa}_{d,\text{init}}$ ,  ${}^d\hat{R}_{b,\text{init}}$ , and  ${}^d\hat{\mathbf{t}}_{b,\text{init}}$  from  $\hat{Z}^c$  via (15), (16), and (17).
  - 5: Numerical-gradient-descent refinement: perform gradient descent on (6) with respect to  $\Theta$ . Return the final  $\hat{\Theta}$  as  $\{{}^b\hat{\delta}_{T,d}, {}^d\hat{R}_b, {}^d\hat{\mathbf{t}}_b, \hat{\kappa}_d\}$ .
- 

along that axis weakly observable. The clock offset  ${}^b\delta_{T,d}$  is mainly informed by translational/rotational accelerations through  $J_{b\delta_{T,d,i}}$ , so nearly constant-speed and constant-turn-rate motions are uninformative for  ${}^b\delta_{T,d}$ , while  $({}^dR_b, \kappa_d)$  benefit from large, direction-rich  $y_i({}^b\delta_{T,d})$ . Accordingly, we recommend a multi-regime calibration trajectory: (a) low angular rate with deliberately varying speed/direction to excite  ${}^b\delta_{T,d}$  while suppressing lever-arm coupling, and (b) sufficiently rich multi-axis rotation with nonzero translation to excite  ${}^d\mathbf{t}_b$  and keep  $y_i({}^b\delta_{T,d})$  informative for  ${}^dR_b$  and  $\kappa_d$ .

## IV. NUMERICAL ANALYSIS

This section presents Monte Carlo simulation results that evaluate the proposed UIC under a camera-DVL calibration problem and quantify how spatiotemporal parameters and motion excitation affect DVL calibration and fusion.

### A. Simulation setup and evaluation protocol

Each trial synthesizes a bounded 3D SE(3) trajectory together with three measurement streams: (i) absolute poses (Abs) available only in a “calibration zone” and under low-speed/low-turn-rate gating, simulating camera visualizing a calibration board. (ii) relative poses (VO) at a fixed rate simulating visual odometry, and (iii) DVL body-frame linear velocities. The ground-truth calibration parameters  $\Theta^o = \{\delta^o, \kappa_d^o, {}^dR_b^o, {}^d\mathbf{t}_b^o\}$  are fixed across MC runs.

### B. UIC against robust baselines

We evaluate both calibration accuracy and downstream trajectory impact in this simulation. Calibration accuracy: Over  $n_{\text{MC}} = 50$  trials, we estimate  $\hat{\Theta}$  and report the errors in  $\delta[\text{s}]$ ,  $\kappa_d$ ,  ${}^dR_b[^\circ]$ , and  ${}^d\mathbf{t}_b[\text{m}]$ . Trajectory

#### D. 迭代改进

从顺序初始化  $\hat{\Theta}$  开始，我们通过问题 (6) 的局部迭代优化来精炼  $\Theta$ 。旋转通过指数映射扰动在  $\text{SO}(3)$  上更新，DVL 比例尺重新参数化以强制  $\kappa_d > 0$ ：

$${}^dR_b \leftarrow \exp(\phi^\wedge) {}^dR_b, \quad \phi \in \mathbb{R}^3, \quad (18)$$

$$\kappa_d = \exp(s_d), \quad s_d \in \mathbb{R}. \quad (19)$$

我们应用数值梯度下降来优化参数。

#### E. 轨迹设计

$\Theta$  的校准精度在很大程度上取决于运动激励，这可以通过由公式 (5) 中 DVL 速度残差模型引入的 Fisher 信息矩阵 (FIM) 来表征。重要的是，预测的机体坐标系运动  $\hat{v}_b(\cdot)$  和  $\hat{\omega}_b(\cdot)$  是从连续时间高斯过程 (GP) 获得的，因此带有不确定性。为了反映这一点，我们使用了一个有效残差协方差  $Q_e$ ，它不仅捕捉了 DVL 测量噪声的内在特性，还捕捉了由估计的运动信号  $\hat{v}_b, \hat{\omega}_b$  引入的不确定性。

将对齐时间  ${}^bT_{d,i}({}^b\delta_{T,d}) = {}^dT_i + {}^b\delta_{T,d}$  的预测 DVL 速度定义为

$$\bar{v}_{d,i}(\Theta) = \kappa_d {}^dR_b \left( \hat{v}_{b,i}({}^b\delta_{T,d}) + \hat{\omega}_{b,i}({}^b\delta_{T,d}) \times {}^d t_b \right).$$

表示  $r_i(\Theta) \triangleq \bar{v}_{d,i} - \hat{v}_{d,i}(\Theta)$  和  $y_i({}^b\delta_{T,d}) \triangleq \hat{v}_{b,i}({}^b\delta_{T,d}) + \hat{\omega}_{b,i}({}^b\delta_{T,d}) \times {}^d t_b$ 。使用无约束的尺度参数化  $\kappa_d = \exp(s_d)$  和左扰动  ${}^dR_b \leftarrow \exp(\phi^\wedge) {}^dR_b$ ，一阶残差雅可比满足

$$J_{s,i} \triangleq \frac{\partial r_i}{\partial s_d} = -\kappa_d {}^dR_b y_i({}^b\delta_{T,d}),$$

$$J_{\phi,i} \triangleq \frac{\partial r_i}{\partial \phi} = \kappa_d ({}^dR_b y_i({}^b\delta_{T,d}))^\wedge,$$

$$J_{t,i} \triangleq \frac{\partial r_i}{\partial ({}^d t_b)} = -\kappa_d {}^dR_b (\hat{\omega}_{b,i}({}^b\delta_{T,d}))^\wedge,$$

$$J_{\delta,i} \triangleq \frac{\partial r_i}{\partial {}^b\delta_{T,d}} = -\kappa_d {}^dR_b \left( \hat{v}_{b,i}({}^b\delta_{T,d}) + \hat{\omega}_{b,i}({}^b\delta_{T,d}) \times {}^d t_b \right),$$

其中  $(\cdot)^\wedge$  表示在对齐时刻评估的时间导数，在连续时间 GP 模型下总是可以解析获取的。

费舍尔信息矩阵可以写成堆叠形式

$$\mathcal{I}(\Theta) = J^\top Q_e^{-1} J, \quad J \triangleq \begin{bmatrix} J_1 \\ \vdots \\ J_N \end{bmatrix}, \quad J_i \triangleq [J_{\delta,i} \quad J_{s,i} \quad J_{\phi,i} \quad J_{t,i}].$$

这里  $Q_e$  是残差的有效协方差，它汇总了 DVL 测量的不确定性以及由运动估计引起的额外不确定性。

从 FIM 来看，杠杆臂  $t_b$  仅由  $J_{t,i} \propto (\hat{\omega}_{b,i}({}^b\delta_{T,d}))^\wedge$  旋转激励；因此  $\hat{\omega}_{b,i}({}^b\delta_{T,d}) \approx 0$  使  $t_b$  不可观测，并且接近单轴旋转会产生一个病态的  $\mathcal{I}_{tt}$ ，其中包含  $t_b$  的分量

---

#### Algorithm 1 UIC: Sequential Initialization + Gradient-Descent Refinement

---

Input: time-stamped DVL measurements  $\{(\hat{v}_{d,i}, T_{d,i})\}_{i=1}^N$ ; time-stamped base measurements  $\{(\hat{m}_{b,j}, T_{b,j})\}_{j=1}^M$ ; GP kernel  $K(t, t')$  in (3); angular-rate threshold  $\omega_{th}$  for selecting  $\mathcal{D}_o$

Output:  ${}^b\hat{\delta}_{T,d}$ ,  $\hat{\kappa}_d$ ,  ${}^d\hat{R}_b$ ,  ${}^d\hat{t}_b$

- 1: GP reference trajectory: compute  $\hat{X}_b(\cdot)$  with covariances from  $\{\hat{m}_{b,j}\}$  and  $K$  via (7)–(9).
  - 2: Temporal/scale pre-initialization on  $\mathcal{D}_o$ : solve (11) by a grid search over  ${}^b\delta_{T,d}$  and  $\kappa_d$  around 1 only on the low-angular-velocity index set.
  - 3: Bias-corrected stacked extrinsics initialization: using  ${}^b\hat{\delta}_{T,d,init}$ , construct  $\hat{H}$  and  $G$  as in (14), and compute  $\hat{Z}^c$  using all  $\hat{v}_{d,i}$ .
  - 4: Recover scale and extrinsics: compute  $\hat{\kappa}_{d,init}$ ,  ${}^d\hat{R}_{b,init}$ , and  ${}^d\hat{t}_{b,init}$  from  $\hat{Z}^c$  via (15), (16), and (17).
  - 5: Numerical-gradient-descent refinement: perform gradient descent on (6) with respect to  $\Theta$ . Return the final  $\hat{\Theta}$  as  $\{{}^b\hat{\delta}_{T,d}, {}^d\hat{R}_b, {}^d\hat{t}_b, \hat{\kappa}_d\}$ .
- 

沿该轴弱可观测。时钟偏移  ${}^b\delta_{T,d}$  主要通过  $J_{b\delta_{T,d,i}}$  受到平移/旋转加速度的影响，因此近似匀速和恒定转速的运动对  ${}^b\delta_{T,d}$  并无信息，而  $({}^dR_b, \kappa_d)$  则受益于大幅度、方向丰富的  $y_i({}^b\delta_{T,d})$ 。因此，我们建议采用多模式校准轨迹：（a）低角速率并有意改变速度/方向以激发  ${}^b\delta_{T,d}$  的同时抑制杆臂耦合；（b）足够丰富的多轴旋转并伴随非零平移，以激发  $t_b$  并保持  $y_i({}^b\delta_{T,d})$  对  ${}^dR_b$  和  $\kappa_d$  的信息性。

#### IV. 数值分析

本节展示了蒙特卡洛仿真结果，用于评估在相机-DVL 标定问题下提出的 UIC，并量化时空参数和运动激励如何影响 DVL 标定和融合。

##### A. 仿真设置与评估协议

每次试验都会合成一个有界的 3D SE(3) 轨迹，以及三个测量流：(i) 仅在“校准区”内并在低速/低转向率条件下可用的绝对位姿 (Abs)，模拟摄像头观测校准板的情况；(ii) 以固定速率获得的相对位姿 (VO)，模拟视觉里程计；以及 (iii) DVL 机体系线速度。真实的校准参数  $\Theta^o = \{\delta^o, \kappa_d^o, {}^dR_b^o, {}^d t_b^o\}$  在蒙特卡洛运行中保持固定。

##### B. UIC 对抗稳健的基线

我们在此模拟中评估了校准精度和下游轨迹影响。校准精度：在超过  $n_{MC} = 1000$  次试验中，我们估计了  $\hat{\Theta}$  并报告了  $\delta$ [秒]、 $\kappa_d$ 、 ${}^dR_b$ [°] 和  ${}^d t_b$ [米] 的误差。轨迹

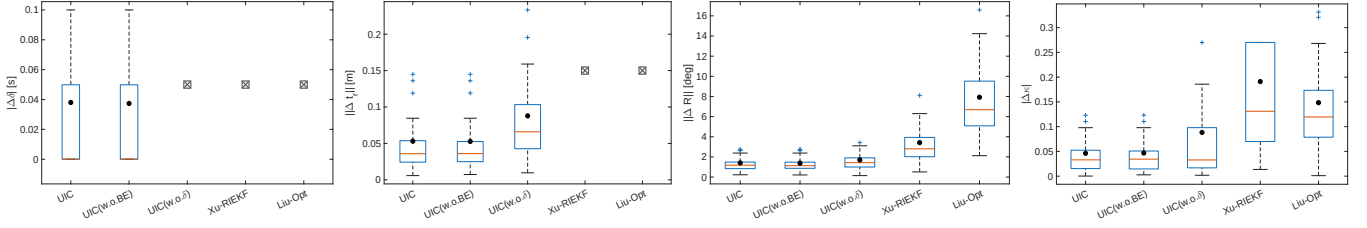


Fig. 2: Calibration Results (Absolute Error distributions of  $\Delta\delta$ ,  $\Delta t$ ,  $\Delta R$ , and  $\Delta\kappa$ ) and RMSE. Boxes show the interquartile range with the median; whiskers indicate the non-outlier range; black dots denote the RMSE.

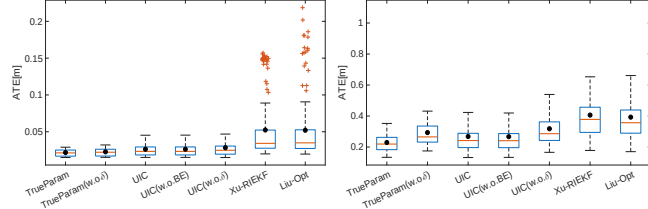


Fig. 3: ATE for simple(left) and aggressive(right) test trajectories

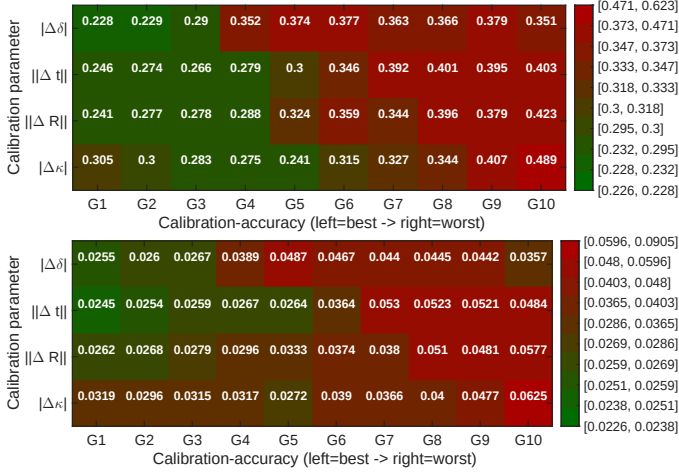


Fig. 4: Calibration accuracy vs Simple(Upper) and Aggressive(Lower) trajectory ATE(Upper)

impact: Each calibrated parameter set is evaluated on two types of independent test trajectories, each of which contains 20 different trajectories. The first type of test trajectories (Simple Trajectories) has slow motions, while the other (Aggressive Trajectories) has rapid movements. We run a GP-InEKF and RTS fusion that incorporates Abs/VO and DVL (with discrete-time offset compensation), and compute the ATE.

To quantify the practical gain of jointly calibrating time and space, we benchmark UIC against two reproduced robust calibration baselines for DVL, Xu-RIEKF [8] and Liu-Opt [10], which estimate only  $(dR_b, \kappa_d)$ ; thus, following their intended usage in our setting, we fix the clock offset  $\delta$  and lever arm  $d_t$  to measured values when running them. We also in-

clude two UIC ablations, UIC(w.o.BE) (without bias-elimination in initialization) and UIC(w.o. $\delta$ ) (disabling temporal calibration by fixing  $\delta = 0$ ), under the same protocol; Figs. 2 and 3 summarize calibration errors and the downstream ATE, respectively. In addition, to explicitly isolate the impact of temporal misalignment on trajectory estimation, we report the baseline TrueParam and TrueParam(w.o. $\delta$ ), where the spatial parameters  $(dR_b, d_t, \kappa_d)$  are set to their ground-truth values while  $\delta$  is set to real value and zero; therefore, any ATE degradation observed in TrueParam(w.o. $\delta$ ) can be attributed to the wrong time alignment rather than imperfect spatial calibration.

Overall, the results show that explicitly estimating a constant  $\delta$  together with  $d_t$  is a key factor for robust and accurate calibration: when  $\delta$  is fixed, the remaining parameters  $(dR_b, \kappa_d, d_t)$  tend to absorb spatiotemporal mismatch, which degrades both calibration quality and trajectory accuracy, whereas UIC yields consistently better calibration and ATE, with the advantage being most pronounced under dynamically rich motions where temporal misalignment is more damaging; consistently, TrueParam(w.o. $\delta$ ) shows only limited ATE change on low-dynamic (Simple) trajectories but a clear degradation on high-dynamic (Aggressive) trajectories, confirming that accurate temporal calibration becomes critical when linear/angular velocities are large. The bias-elimination option provides a modest but consistent improvement for spatial parameters, but its impact is smaller than that of enabling temporal calibration and lever-arm estimation, indicating that these two components dominate this benchmark. Finally, the stratified analysis in Figs. 4 confirms that trajectory accuracy deteriorates as calibration quality worsens, and that the trajectory is generally more sensitive to the clock offset  $\delta$  than to the scale  $\kappa_d$ .

### C. Angular velocity excitation ablation with fixed time offset

This experiment isolates the effect of angular velocity excitation on the spatial DVL-IMU calibration, with an emphasis on the lever arm  $d_t$ . From section III-E, when angular motion is weak, the lever arm becomes weakly observable. To decouple spatial calibration from temporal misalignment, we fix the time offset to the

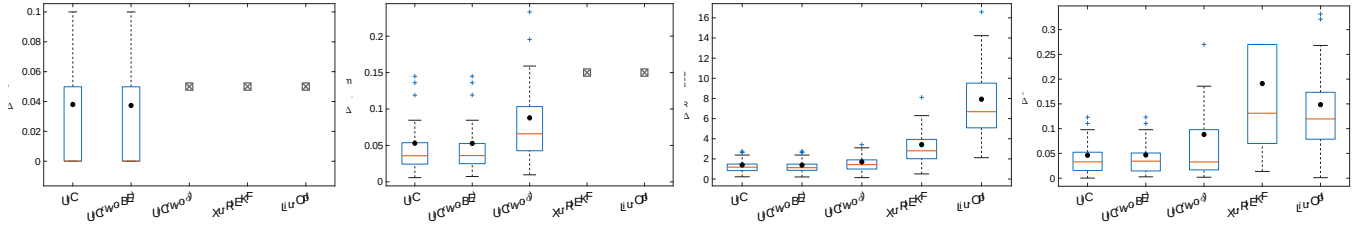


图2: 校准结果 ( $\Delta\delta$ 、 $\Delta t$ 、 $\Delta R$  和  $\Delta\kappa$  的绝对误差分布) 及均方根误差 (RMSE)。箱体显示四分位距及中位数; 须线表示非异常值范围; 黑点表示均方根误差 (RMSE)。

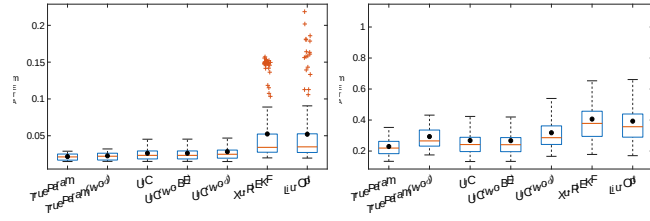


图3: 简单 (左) 和激进 (右) 测试轨迹的 ATE

包括两个UIC消融实验, UIC(w.o.BE) (初始化时不进行偏差消除) 和UIC(w.o. $\delta$ ) (通过固定 $\delta = 0$ 禁用时间校准), 在相同协议下进行; 图2和图3分别总结了校准误差和下游ATE。此外, 为了明确隔离时间错位对轨迹估计的影响, 我们报告了基线TrueParam和TrueParam(w.o. $\delta$ ), 其中空间参数 ( $dR_b, d_t_b, \kappa_d$ ) 设置为其真实值, 而 $\delta$ 则设置为真实值和零; 因此, 在TrueParam(w.o. $\delta$ )中观察到的任何ATE下降都可以归因于错误的时间对齐, 而不是空间校准不完善。

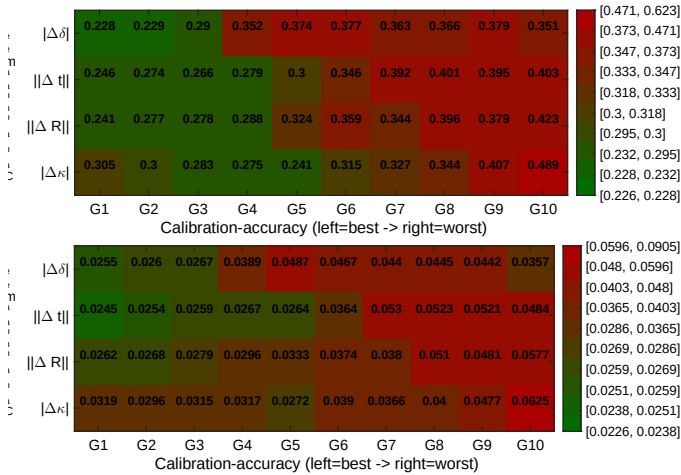


图4: 校准精度与简单 (上) 和激进 (下) 轨迹 ATE (上)

影响: 每组校准的参数都在两种类型的独立测试轨迹上进行评估, 每种类型包含20条不同的轨迹。第一种测试轨迹 (简单轨迹) 运动较慢, 而另一种 (激进轨迹) 运动较快。我们运行一个融合了绝对定位/视觉里程计和多普勒速度测量仪 (带离散时间偏移补偿) 的GP-InEKF和RTS, 并计算ATE。

为了量化联合校准时间和空间的实际收益, 我们将UIC与两个为DVL复现的鲁棒校准基线进行基准测试, 分别是Xu-RIEKF [8]和Liu-Opt [10], 它们仅估计( $dR_b, \kappa_d$ ); 因此, 在我们的设置中按照它们的预期使用方式, 我们在运行它们时将时钟偏移 $\delta$ 和力臂 $d_t_b$ 固定为测量值。我们还在

总体而言, 结果显示, 同时显式估计一个常数 $\delta$ 与 $d_t_b$ 是实现稳健且精确校准的关键因素: 当 $\delta$ 固定时, 其余参数 ( $dR_b, \kappa_d, d_t_b$ ) 倾向于吸收时空不匹配, 这会降低校准质量和轨迹精度, 而UIC则始终能提供更好的校准和ATE, 其优势在动态丰富的运动下最为显著, 因为时间错位的影响更大; 一致地, TrueParam(w.o. $\delta$ )在低动态 (简单) 轨迹上仅显示有限的ATE变化, 但在高动态 (激进) 轨迹上表现出明显的退化, 这确认了当线速度/角速度较大时, 准确的时间校准变得至关重要。偏差消除选项对空间参数提供了适度但稳定的改进, 但其影响小于启用时间校准和杠杆臂估计, 这表明这两个组件在中起主导作用。

### C. 具有固定时间偏移的角速度激励消融

本实验隔离了角速度激励对空间 DVL-IMU 校准的影响, 重点是力臂  $d_t_b$ 。根据第三节 E 部分, 当角运动较弱时, 力臂变得弱可观。为了将空间校准与时间错位解耦, 我们将时间偏移固定为

| Case      | $T$ [s] | $\ \omega_b\ $ [rad/s] | $\Delta\kappa_d$ | $\Delta R$ [deg] | $\Delta t$ [m] | runtime [s] |
|-----------|---------|------------------------|------------------|------------------|----------------|-------------|
| HIGH_T100 | 100     | 0.602                  | 0.0683           | 1.563            | 0.0976         | 0.510       |
| LOW_T100  | 100     | 0.381                  | 0.0596           | 1.567            | 0.1904         | 0.507       |
| LOW_T2000 | 2000    | 0.393                  | 0.0900           | 0.530            | 0.1325         | 18.35       |

TABLE I: Summary for the angular velocity excitation ablation.

ground truth and disable its estimation in UIC, i.e.,  $\delta = \delta^o = 0.05$  s. Therefore the time-offset error is identically zero ( $\text{RMSE}(\Delta\delta) = 0$ ) by construction, and UIC estimates only the DVL scale factor, rotation, and lever arm.

We compare three trajectory scenarios over  $n_{\text{MC}} = 50$  Monte Carlo runs: (i) HIGH\_T100: high angular velocity excitation,  $T = 100$  s; (ii) LOW\_T100: low angular velocity excitation,  $T = 100$  s; (iii) LOW\_T2000: low angular velocity excitation,  $T = 2000$  s. As a scalar proxy for excitation strength, we report the mean ground-truth angular-speed magnitude  $\|\omega_b\|$ .

Table I show that lever-arm accuracy is strongly excitation-limited. For the same duration  $T = 100$  s, increasing angular-rate excitation reduces the lever-arm RMSE. In contrast, the rotation error remains around  $\approx 1.56$  deg for both 100 s cases, suggesting that in this regime  ${}^dR_b$  is less sensitive to short-horizon excitation amplitude than  ${}^d\mathbf{t}_b$ . Extending a low-excitation trajectory to  $T = 2000$  s improves the rotation error substantially and also improves lever-arm accuracy, but it increases computation time markedly.

## V. Real-World Experiment

We conducted a real-world experiment to evaluate our calibration method. We recorded our calibration trajectories in a  $4.5 \times 10 \times 2$  m pool. An A50 DVL (around 10 Hz) is mounted on an underwater robot together with a forward-looking stereo camera (10 Hz). Fig. 6 shows the setting of our experiment. The stereo camera provides absolute pose measurements by observing an AprilTag board affixed to the pool wall. We calibrate the camera-DVL extrinsics  ${}^dR_c, {}^d\mathbf{t}_c$ , their clock offset  ${}^c\delta_{T,d}$ , and the DVL scale factor  $\kappa_d$ .

We apply GP with WNOJ [22] prior to estimating trajectories and manually set the kernel  $Q_c$  as  $\text{diag}([100, 100, 100, 100, 100, 100])$ . To evaluate the calibration performance, we construct an odometry that integrates the DVL linear velocity measurements with the stereo camera’s absolute orientation measurements as follows. Let  ${}^wR_c(T_c)$  denote the camera orientation in the world frame at time  $T_c$ . Over a short interval  $[T_{c,1}, T_{c,2}]$  between two camera measurements, since the DVL has the same frequency as the camera, there exist two DVL measurements  $\hat{v}_{d,1}$  and  $\hat{v}_{d,2}$  at times  $T_{d,1} + {}^c\delta_{T,d}$  and  $T_{d,2} + {}^c\delta_{T,d}$ , which satisfy  $T_{d,1} + {}^c\delta_{T,d} \leq T_{c,1}$  and  $T_{d,2} + {}^c\delta_{T,d} \in [T_{c,1}, T_{c,2}]$ . Then, the average velocity in this interval can be approximated

by  $v_d = \frac{T_{d,2} + {}^c\delta_{T,d} - T_{c,1}}{\Delta_T} \hat{v}_{d,1} + \frac{T_{c,2} - T_{d,2} - {}^c\delta_{T,d}}{\Delta_T} \hat{v}_{d,2}$ , where  $\Delta_T = T_{c,2} - T_{c,1}$ . Finally, the world-frame camera position update over this interval is approximated by  ${}^w\mathbf{p}_2 \approx {}^w\mathbf{p}_1 + \kappa_d^{-1} {}^wR_c(T_{c,1}) {}^cR_d v_d \Delta_T + ({}^wR_c(T_{c,1}) - {}^wR_c(T_{c,2})) {}^d\mathbf{t}_c$ , where  $\kappa_d^{-1} {}^wR_c(T_{c,1}) {}^cR_d v_d \Delta_T$  and  $({}^wR_c(T_{c,1}) - {}^wR_c(T_{c,2})) {}^d\mathbf{t}_c$  denote the updates brought by robot translation and rotation, respectively.

We compare the above estimated position-only trajectory against the reference trajectory derived from the stereo camera’s absolute-position measurements from the AprilTag board. Absolute Trajectory Error (ATE) and Relative Position Error (RPE) are selected as evaluation metrics. We evaluate five methods and compare their performance:

- UIC: our UIC method in Algorithm 1.
- GD (init. 1): gradient decent over Problem (6) initialized with CAD-derived extrinsic parameters.
- GD (init. 2): gradient decent over Problem (6) initialized with  ${}^dR_b = \mathbf{I}_3$ ,  ${}^d\mathbf{t}_b = \mathbf{0}$ .
- Manual w.  $\delta_T$ : use manually measured  ${}^dR_b$  and  ${}^d\mathbf{t}_b$  obtained from the CAD assembly model, and compensate for the clock offset estimated by UIC.
- Manual w.o.  $\delta_T$ : use manually measured  ${}^dR_b$  and  ${}^d\mathbf{t}_b$  obtained from the CAD assembly model, with the clock offset fixed at 0.

We recorded three mild-motion trajectories, denoted as A, B, and C, under the same pool and camera-visibility constraints. Each trajectory is used for calibration and evaluated on the other two, yielding six cross-trajectory pairs; the results are summarized in Table II, with representative trajectories shown in Fig. 5. Trajectory C is generally more challenging for the downstream odometry: nearly all methods show larger ATE when evaluated on C, while UIC remains comparable to well-initialized GD and achieves the best average ATE and RPE without manual initialization. Poorly initialized GD is unstable, and UIC outperforms both manual base-lines. The small gap between manual calibration with and without time-offset compensation is consistent with the mild motions; simulations show a larger temporal effect under aggressive dynamics.

Limitations: The lever arm  ${}^d\mathbf{t}_c$  is weakly observable in the pool experiment because keeping the AprilTag board within the camera’s field of view limits rotational excitation; consequently, the  $\omega \times {}^d\mathbf{t}_c$  contribution in Fig. 7 is negligible. This is primarily an excitation limitation: richer multi-axis rotation, enabled by a wider-view or multi-target reference setup, is the preferred remedy. Priors may stabilize estimation but cannot replace missing excitation.

## VI. CONCLUSION

In this paper, we formalized DVL spatiotemporal calibration as an error-in-variables nonlinear least-squares problem under a continuous-time GP motion model, and introduced a UIC framework centered on a sequential,

| Case      | $T$ [s] | $\ \omega_b\ $ [rad/s] | $\Delta\kappa_d$ | $\Delta R$ [deg] | $\Delta t$ [m] | runtime [s] |
|-----------|---------|------------------------|------------------|------------------|----------------|-------------|
| HIGH_T100 | 100     | 0.602                  | 0.0683           | 1.563            | 0.0976         | 0.510       |
| LOW_T100  | 100     | 0.381                  | 0.0596           | 1.567            | 0.1904         | 0.507       |
| LOW_T2000 | 2000    | 0.393                  | 0.0900           | 0.530            | 0.1325         | 18.35       |

表 I: 角速度激励消融的总结。

在UIC中使用真实值并禁用其估计，即  $\delta = \delta^o = 0.05$  s。因此，时间偏移误差在构造上完全为零（RMSE( $\Delta\delta$ ) = 0），UIC仅估计DVL尺度因子、旋转和杆臂。

我们在  $n_{MC} =$  次蒙特卡洛试验中比较了三种轨迹方案：(i) HIGH\_T100: 高角速度激励， $T = 100$  秒；(ii) LOW\_T100: 低角速度激励， $T = 100$  秒；(iii) LOW\_T2000: 低角速度激励， $T = 2000$  秒。作为激励强度的标量代理，我们报告了真实角速度大小的均值  $\|\omega_b\|$ 。

表 I 显示，杠杆臂精度在很大程度上受激励限制。对于相同的持续时间  $T = 100$  秒，增加角速度激励会降低杠杆臂的均方根误差 (RMSE)。相比之下，旋转误差在两个 100 秒的情况下都保持在  $\approx 1.56$  度左右，这表明在此范围内， ${}^dR_b$  对短期激励幅度的敏感性低于  $q_b$ 。将低激励轨迹延长至  $T = 2000$  秒可以显著改善旋转误差，也能提高杠杆臂精度，但会显著增加计算时间。

## 五、现实世界实验

我们进行了一个真实环境实验来评估我们的校准方法。我们在一个  $4.5 \times 10 \times 2$  米的游泳池中记录了我们的校准轨迹。一台 A50 DVL (约 10 Hz) 安装在水下机器人上，同时配备了一台前视立体相机 (10 Hz)。图 6 显示了我们实验的设置。立体相机通过观察固定在泳池墙上的 AprilTag 板提供绝对位姿测量。我们对相机-DVL 外参  ${}^dR_c, q_c$ 、它们的时钟偏移  ${}^c\delta_{T,d}$  以及 DVL 比例因子  $\kappa_d$  进行校准。

我们在估计轨迹之前应用带有 WNOJ [22] 的 GP，并手动将核  $Q_c$  设置为  $\text{diag}([100, 100, 100, 100, 100, 100])$ 。为了评估校准性能，我们构建了一个里程计，将 DVL 线速度测量与立体相机的绝对姿态测量结合如下。设  ${}^wR_c(T_c)$  表示时间  $T_c$  时相机在世界坐标系中的姿态。在两次相机测量之间的短时间间隔  $[T_{c,1}, T_{c,2}]$  内，由于 DVL 与相机具有相同的频率，因此在时间  $T_{d,1} + {}^c\delta_{T,d}$  和  $T_{d,2} + {}^c\delta_{T,d}$  存在两个 DVL 测量  $\hat{v}_{d,1}$  和  $\hat{v}_{d,2}$ ，它们满足  $T_{d,1} + {}^c\delta_{T,d} \leq T_{c,1}$  和  $T_{d,2} + {}^c\delta_{T,d} \in [T_{c,1}, T_{c,2}]$ 。然后，该时间间隔内的平均速度可以近似为

通过 
$$v_d = \frac{T_{d,2} + {}^c\delta_{T,d} - T_{c,1}}{T_{c,2} - T_{c,1}} \hat{v}_{d,1} + \frac{T_{c,2} - T_{d,2} - {}^c\delta_{T,d}}{T_{c,2} - T_{c,1}} \hat{v}_{d,2},$$
 其中  $\Delta T = T_{c,2} - T_{c,1}$ 。最后，该时间间隔内的世界坐标系相机位置更新大约由 
$${}^wp_2 \approx {}^wp_1 + \kappa_d^{-1} {}^wR_c(T_{c,1}) {}^cR_d v_d \Delta T + ({}^wR_c(T_{c,1}) - {}^wR_c(T_{c,2})) q_c$$
 近似表示，其中  $\kappa_d^{-1} {}^wR_c(T_{c,1}) {}^cR_d v_d \Delta T$  和  $({}^wR_c(T_{c,1}) - {}^wR_c(T_{c,2})) q_c$  分别表示机器人平移和旋转带来的更新。

我们将上述仅基于位置的估计轨迹与使用立体相机从 AprilTag 板测得的绝对位置生成的参考轨迹进行比较。选择绝对轨迹误差 (ATE) 和相对位置误差 (RPE) 作为评估指标。我们评估五种方法并比较它们的性能：

- UIC: 我们在算法1中的UIC方法。
- GD (初始值1): 在问题 (6) 上进行梯度下降，初始值使用 CAD 派生的外参。
- GD (初始化 2): 在问题 (6) 上进行梯度下降，初始化为  ${}^dR_b = \mathbf{I}_3$ ,  $q_b = \mathbf{0}$ 。
- 手动 w.  $\delta_T$ : 使用从 CAD 装配模型中手动测量的  ${}^dR_b$  和  $q_b$ ，并补偿 UIC 估算的时钟偏移。
- 手动 w.o.  $\delta_T$ : 使用从 CAD 装配模型手动测量的  ${}^dR_b$  和  $q_b$ ，时钟偏移固定为 0。

我们记录了三条轻度运动轨迹，分别标记为 A、B 和 C，在相同的池和相机可视性约束下进行。每条轨迹都用于校准，并在其他两条轨迹上进行评估，产生六组交叉轨迹对；结果总结在表 II 中，并在图 5 中展示了代表性轨迹。对于下游里程计来说，轨迹 C 通常更具挑战性：几乎所有方法在 C 上的评估中均显示出更大的 ATE，而 UIC 则保持与良好初始化的 GD 可比，并在无需人工初始化的情况下实现了最佳平均 ATE 和 RPE。初始化不佳的 GD 不稳定，而 UIC 优于两种人工基线。带时间偏移补偿与不带补偿的人工校准之间的差距较小，这与轻度运动一致；仿真显示在激烈动力学下时间效应更明显。

局限性：在池实验中，杠杆臂  $q_c$  的可观测性较弱，因为为了保持 AprilTag 板在相机的视野内，旋转激励受到限制；因此，图 7 中  $\omega \times q_c$  的贡献可以忽略不计。这主要是激励的局限性：通过更广视角或多目标参考设置实现更丰富的多轴旋转，是首选的解决办法。先验信息可以稳定估计，但无法替代缺失的激励。

## 六、结论

在本文中，我们将DVL时空校准形式化为基于连续时间GP运动模型的变量误差非线性最小二乘问题，并引入了一个以序列为中心的UIC框架，



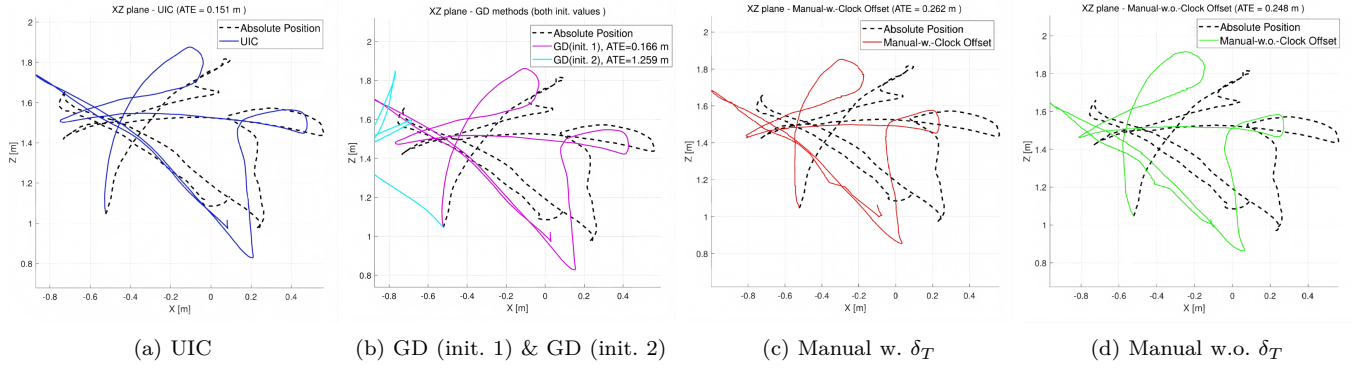


Fig. 5: Trajectory A estimation using the parameters calibrated from trajectory B.

TABLE II: Cross-trajectory ATE and RPE. Each column pair lists the evaluation trajectory under the calibration trajectory shown above. **Bold blue** and **light blue** denote the best and second-best results. Averages are arithmetic means over the six pairs; percentages are relative to UIC and are computed from the listed per-pair values before rounding the averages.

| Calibration        |        | Traj A       |              |              |              | Traj B       |              |              |               | Traj C       |              |              |              | Average      |              |
|--------------------|--------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|---------------|--------------|--------------|--------------|--------------|--------------|--------------|
| Evaluate           |        | Traj B       |              | Traj C       |              | Traj A       |              | Traj C       |               | Traj A       |              | Traj B       |              | Comparison   |              |
| Method             | Result | ATE          | RPE          | ATE          | RPE          | ATE          | RPE          | ATE          | RPE           | ATE          | RPE          | ATE          | RPE          | ATE          | RPE          |
| UIC                | Meter  | <b>0.348</b> | <b>0.107</b> | <b>0.621</b> | 0.156        | <b>0.151</b> | <b>0.070</b> | <b>0.538</b> | <b>0.141</b>  | <b>0.209</b> | <b>0.067</b> | <b>0.230</b> | <b>0.071</b> | <b>0.350</b> | <b>0.102</b> |
| GD                 | Meter  | 0.368        | <b>0.107</b> | <b>0.606</b> | <b>0.154</b> | <b>0.166</b> | <b>0.070</b> | <b>0.529</b> | 0.142         | <b>0.231</b> | <b>0.070</b> | <b>0.237</b> | <b>0.072</b> | <b>0.356</b> | <b>0.103</b> |
| (init. 1)          | %      | 105.7        | <b>100</b>   | <b>97.58</b> | <b>98.72</b> | <b>109.9</b> | <b>100</b>   | <b>98.3</b>  | <b>100.71</b> | <b>110.5</b> | <b>104.5</b> | <b>103.0</b> | <b>101.4</b> | <b>101.9</b> | <b>100.5</b> |
| GD                 | Meter  | <b>0.367</b> | <b>0.107</b> | 0.993        | <b>0.144</b> | 1.259        | 0.201        | 0.942        | <b>0.138</b>  | 1.158        | 0.196        | 0.750        | 0.102        | 0.912        | 0.148        |
| (init. 2)          | %      | <b>105.5</b> | <b>100</b>   | 159.9        | <b>92.31</b> | 833.8        | 287.1        | 175.1        | <b>97.9</b>   | 554.1        | 292.5        | 326.1        | 143.7        | 260.8        | 145.1        |
| Manual             | Meter  | 0.378        | <b>0.108</b> | <b>0.621</b> | 0.158        | 0.262        | <b>0.085</b> | 0.607        | 0.157         | 0.262        | 0.085        | 0.390        | 0.109        | 0.420        | 0.117        |
| (w. $\delta_T$ )   | %      | 108.6        | <b>100.9</b> | <b>100</b>   | 101.3        | 173.5        | <b>121.4</b> | 112.8        | 111.3         | 125.4        | 126.9        | 169.6        | 153.5        | 120.2        | 114.7        |
| Manual             | Meter  | 0.388        | <b>0.108</b> | 0.632        | 0.158        | 0.248        | 0.088        | 0.632        | 0.158         | 0.248        | 0.088        | 0.388        | 0.108        | 0.423        | 0.118        |
| (w.o. $\delta_T$ ) | %      | 111.5        | <b>100.9</b> | 101.8        | 101.3        | 164.2        | 125.7        | 117.5        | 112.1         | 118.7        | 131.3        | 168.7        | 152.1        | 120.9        | 115.7        |

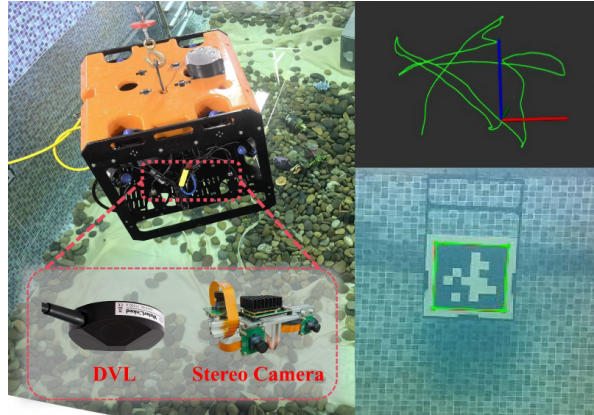


Fig. 6: Water pool experiments. Left: Underwater robot; bottom right: AprilTag board; upper right: robot trajectory.

data-driven initialization procedure with gradient descent refinement. Under idealized conditions, we characterized the large-sample behavior of the proposed initialization estimators. Our simulations show that lever-arm estimation is excitation-limited, and the pool experiment validates the practicality of the proposed initializer under our platform constraints. Monte Carlo ablation studies further highlight that explicitly calibrating the clock offset and executing sufficiently rotationally exciting trajectories are critical for reliable spatiotemporal cal-

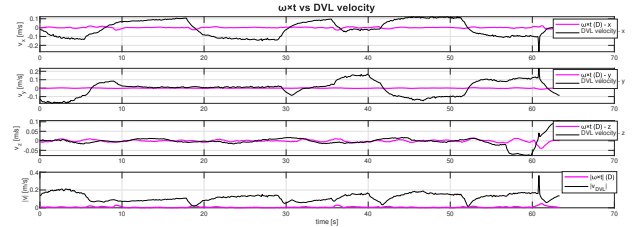


Fig. 7: DVL velocity versus the lever-arm-induced term  $\omega \times d_c$ .

ibration in finite data.

## Acknowledgement

This work was supported in part by 1+1+1 CUHK-CUHK(SZ)-GDSTC Joint Collaboration Fund, JCYJ20241202124010014.

## References

- [1] L. Zhao, M. Zhou, and B. Loose, “Tightly-coupled visual-dvl- inertial odometry for robot-based ice-water boundary exploration,” in 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023, pp. 7127–7134.
- [2] Y. Huang, P. Li, S. Yan, Y. Ou, Z. Wu, M. Tan, and J. Yu, “Tightly-coupled visual-dvl fusion for accurate localization of underwater robots,” in 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2023, pp. 8090–8095.

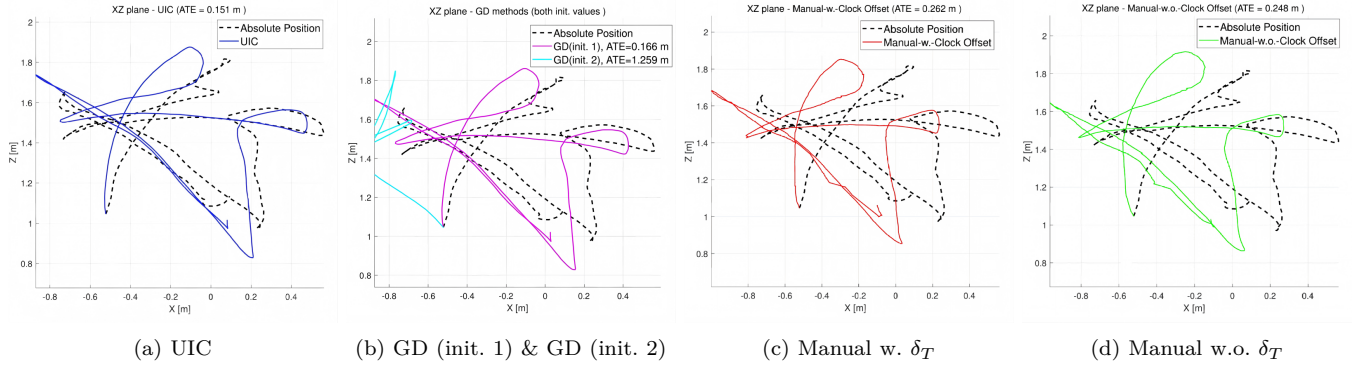


图5: 轨迹 使用从轨迹校准的参数估计

表 II: 跨轨迹 ATE 和 RPE。

每对列显示在上方所示校准轨迹下的评估轨迹。粗体蓝色和浅蓝色分别表示最佳和次佳结果。平均值是六对的算术平均数；百分比是相对于 UIC 的，并根据列出的每对值计算，平均值取整之前计算。

| Calibration               |        | Traj A |       |        |       | Traj B |       |        |        | Traj C |       |        |       | Average    |       |
|---------------------------|--------|--------|-------|--------|-------|--------|-------|--------|--------|--------|-------|--------|-------|------------|-------|
| Evaluate                  |        | Traj B |       | Traj C |       | Traj A |       | Traj C |        | Traj A |       | Traj B |       | Comparison |       |
| Method                    | Result | ATE    | RPE   | ATE    | RPE   | ATE    | RPE   | ATE    | RPE    | ATE    | RPE   | ATE    | RPE   | ATE        | RPE   |
| UIC                       | Meter  | 0.348  | 0.107 | 0.621  | 0.156 | 0.151  | 0.070 | 0.538  | 0.141  | 0.209  | 0.067 | 0.230  | 0.071 | 0.350      | 0.102 |
| GD                        | Meter  | 0.368  | 0.107 | 0.606  | 0.154 | 0.166  | 0.070 | 0.529  | 0.142  | 0.231  | 0.070 | 0.237  | 0.072 | 0.356      | 0.103 |
| GD (init. 1)              | %      | 105.7  | 100   | 97.58  | 98.72 | 109.9  | 100   | 98.3   | 100.71 | 110.5  | 104.5 | 103.0  | 101.4 | 101.9      | 100.5 |
| GD (init. 2)              | Meter  | 0.367  | 0.107 | 0.993  | 0.144 | 1.259  | 0.201 | 0.942  | 0.138  | 1.158  | 0.196 | 0.750  | 0.102 | 0.912      | 0.148 |
|                           | %      | 105.5  | 100   | 159.9  | 92.31 | 833.8  | 287.1 | 175.1  | 97.9   | 554.1  | 292.5 | 326.1  | 143.7 | 260.8      | 145.1 |
| Manual (w. $\delta_T$ )   | Meter  | 0.378  | 0.108 | 0.621  | 0.158 | 0.262  | 0.085 | 0.607  | 0.157  | 0.262  | 0.085 | 0.390  | 0.109 | 0.420      | 0.117 |
|                           | %      | 108.6  | 100.9 | 100    | 101.3 | 173.5  | 121.4 | 112.8  | 111.3  | 125.4  | 126.9 | 169.6  | 153.5 | 120.2      | 114.7 |
| Manual (w.o. $\delta_T$ ) | Meter  | 0.388  | 0.108 | 0.632  | 0.158 | 0.248  | 0.088 | 0.632  | 0.158  | 0.248  | 0.088 | 0.388  | 0.108 | 0.423      | 0.118 |
|                           | %      | 111.5  | 100.9 | 101.8  | 101.3 | 164.2  | 125.7 | 117.5  | 112.1  | 118.7  | 131.3 | 168.7  | 152.1 | 120.9      | 115.7 |

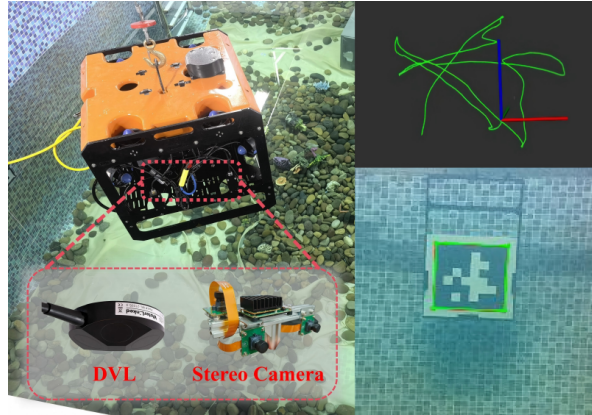


图6: 水池实验。左: 水下机器人; 右下: AprilTag 标板; 右上: 机器人轨迹。

基于数据的初始化过程结合梯度下降优化。在理想化条件下，我们刻画了所提出的初始化估计量的大样本行为。我们的仿真表明，杠杆臂估计受激励限制，而池实验验证了在我们的平台约束下所提出的初始化器的实用性。蒙特卡洛消融研究进一步强调，明确校准时钟偏移并执行足够旋转激励的轨迹对于可靠的时空校准至关重要。

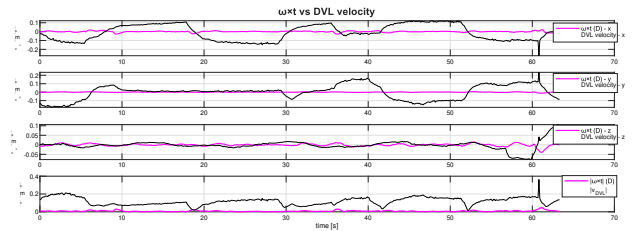


图7: DVL速度与由力臂引起的项  $\omega \times t_c$  的关系。

有限数据中的振动。

## 致谢

本工作部分得到了 1+1+1 香港中文大学-香港中文大学 (深圳) -广东省科技厅联合合作基金 JCYJ202412021 24010014 的支持。

## 参考文献

- [1] L. 赵, M. 周, 和 B. Loose, “用于基于机器人冰水边界探测的紧耦合视觉-DVL-惯性里程计,” 收录于 2023 年 IEEE/RSJ 国际智能机器人与系统会议 (IROS), 2023, 第 7127–7134 页。
- [2] Y. 黄, P. 李, S. 闫, Y. 欧, Z. 吴, M. 谭, 和 J. 余, “用于水下机器人精确定位的紧耦合视觉-DVL 融合,” 收录于 2023 年 IEEE/RSJ 国际智能机器人与系统会议 (IROS), 2023, 第 8090–8095 页。

- [3] S. Xu, K. Zhang, Z. Hong, Y. Liu, and S. Wang, "Diso: Direct imaging sonar odometry," in 2024 IEEE International Conference on Robotics and Automation (ICRA), 2024, pp. 8573–8579.
- [4] J. McConnell, F. Chen, and B. Englot, "Overhead image factors for underwater sonar-based slam," IEEE Robotics and Automation Letters, vol. 7, no. 2, pp. 4901–4908, 2022.
- [5] S. E. Webster, L. L. Whitcomb, and R. M. Eustice, "Preliminary results in decentralized estimation for single-beacon acoustic underwater navigation," in Robotics: Science and Systems, vol. 10, 2010.
- [6] E. Westman and M. Kaess, "Underwater apriltag slam and calibration for high precision robot localization," in tech. rep. Carnegie Mellon University Pittsburgh, PA, 2018.
- [7] S. Xu, T. Luczynski, J. S. Willners, Z. Hong, K. Zhang, Y. R. Petillot, and S. Wang, "Underwater visual acoustic slam with extrinsic calibration," in 2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2021, pp. 7647–7652.
- [8] B. Xu and Y. Guo, "A novel dvl calibration method based on robust invariant extended kalman filter," IEEE Transactions on Vehicular Technology, vol. 71, no. 9, pp. 9422–9434, 2022.
- [9] B. Xu, L. Wang, S. Li, and J. Zhang, "A novel calibration method of sins/dvl integration navigation system based on quaternion," IEEE Sensors Journal, vol. 20, no. 16, pp. 9567–9580, 2020.
- [10] S. Liu, T. Zhang, and Y. Zhu, "A gnss aided calibration method for dvl error based on the optimal-request," IEEE Sensors Journal, vol. 22, no. 22, pp. 21 899–21 910, 2022.
- [11] B. Wang, J. Liu, Z. Deng, and M. Fu, "A model-free calibration method of inertial navigation system and doppler sensors," IEEE Sensors Journal, vol. 21, no. 2, pp. 2219–2229, 2021.
- [12] P. Furgale, J. Rehder, and R. Siegwart, "Unified temporal and spatial calibration for multi-sensor systems," in 2013 IEEE/RSJ International Conference on Intelligent Robots and Systems, 2013, pp. 1280–1286.
- [13] T. Qin and S. Shen, "Online temporal calibration for monocular visual-inertial systems," in 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2018, pp. 3662–3669.
- [14] J. Peršić, L. Petrović, I. Marković, and I. Petrović, "Spatiotemporal multisensor calibration via gaussian processes moving target tracking," IEEE Transactions on Robotics, vol. 37, no. 5, pp. 1401–1415, 2021.
- [15] M. Li and A. I. Mourikis, "3-d motion estimation and online temporal calibration for camera-imu systems," in 2013 IEEE International Conference on Robotics and Automation, 2013, pp. 5709–5716.
- [16] J. Rehder, R. Siegwart, and P. Furgale, "A general approach to spatiotemporal calibration in multisensor systems," IEEE Transactions on Robotics, vol. 32, no. 2, pp. 383–398, 2016.
- [17] C. K. Williams and C. E. Rasmussen, Gaussian processes for machine learning. MIT press Cambridge, MA, 2006, vol. 2, no. 3.
- [18] A. Wald, "The fitting of straight lines if both variables are subject to error," The annals of mathematical statistics, vol. 11, no. 3, pp. 284–300, 1940.
- [19] B.-Q. Mu and H.-F. Chen, "Recursive identification of multi-input multi-output errors-in-variables hammerstein systems," IEEE Transactions on Automatic Control, vol. 60, no. 3, pp. 843–849, 2014.
- [20] O. Reiersøl, "Identifiability of a linear relation between variables which are subject to error," Econometrica: Journal of the Econometric Society, pp. 375–389, 1950.
- [21] C. H. Tong, P. Furgale, and T. D. Barfoot, "Gaussian process gauss-newton for non-parametric simultaneous localization and mapping," The International Journal of Robotics Research, vol. 32, no. 5, pp. 507–525, 2013. [Online]. Available: <https://doi.org/10.1177/0278364913478672>
- [22] T. Y. Tang, D. J. Yoon, and T. D. Barfoot, "A white-noise-on-jerk motion prior for continuous-time trajectory estimation on  $se(3)$ ," IEEE Robotics and Automation Letters, vol. 4, no. 2, pp. 594–601, 2019.

[3] S. Xu, K. Zhang, Z. Hong, Y. Liu 和 S. Wang, “Diso: 直接成像声呐里程计,” 载于 2024 年 IEEE 国际机器人与自动化会议 (ICRA), 2024, 第 8573–8579 页. [4] J. McConnell, F. Chen 和 B. Englot, “基于水下声呐的 SLAM 之俯视图因子,” IEEE 机器人与自动化快报, 第 7 卷, 第 2 期, 第 4901–4908 页, 2022 年. [5] S. E. Webster, L. L. Whitcomb 和 R. M. Eustice, “单信标声学水下导航的分散估计初步结果,” 载于 *Robotics: Science and Systems*, 第 10 卷, 2010 年. [6] E. Westman 和 M. Kaess, “水下 AprilTag SLAM 及用于高精度机器人定位的校准,” 技术报告, 卡内基梅隆大学, 匹兹堡, PA, 2018 年. [7] S. Xu, T. Luczynski, J. S. Willner, Z. Hong, K. Zhang, Y. R. Petillot 和 S. Wang, “带外参校准的水下视觉声学 SLAM,” 载于 2021 年 IEEE/RSJ 国际智能机器人与系统会议 (IROS), IEEE, 2021, 第 7647–7652 页.